

Microbial cause of Skin and soft tissue infection

Skin and Soft-Tissue Infections

IMPETIGO, ABSCESES, CELLULITIS, AND ERYSIPELA

Objectives

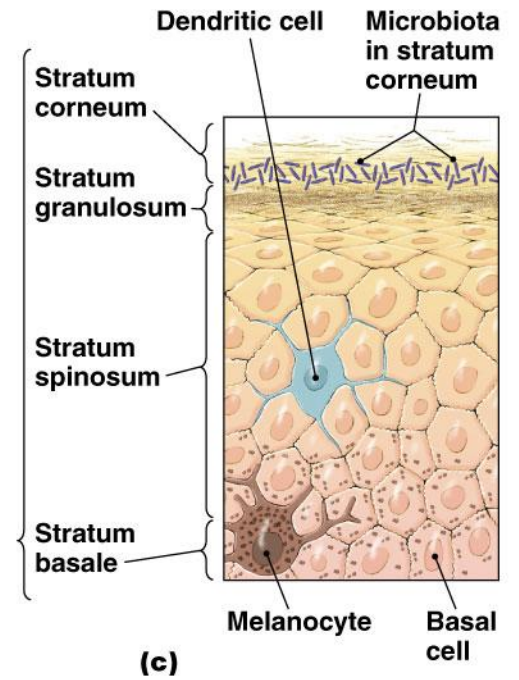
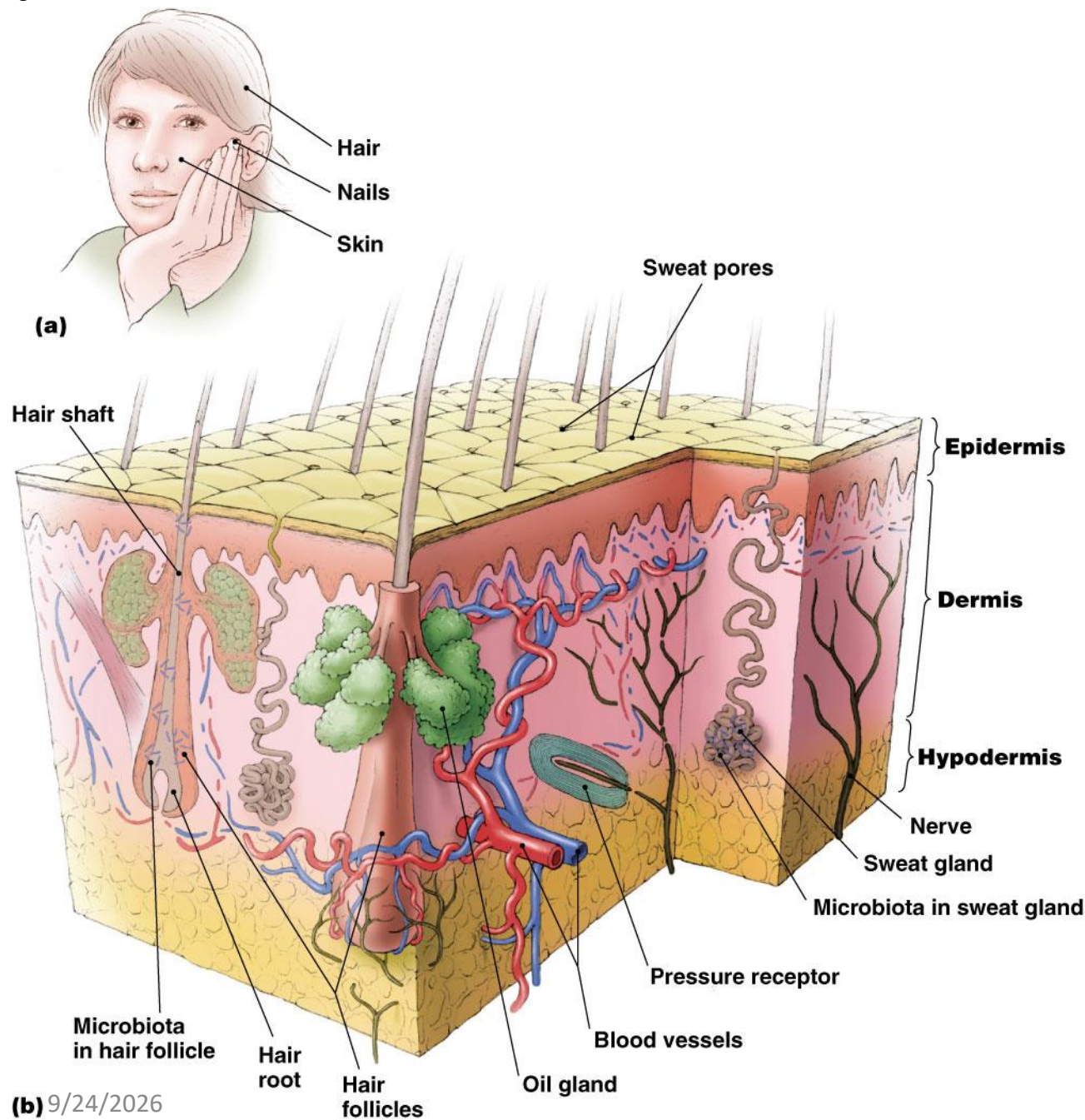
At the end of this session the learner will be able to:

- ✓ Describe the anatomical structure of skin and soft tissues.
- ✓ Differentiate the various types of skin and soft tissue infections and their clinical presentation.
- ✓ Name bacteria, fungus, viruses and parasite commonly involved in skin and soft tissue infections
- ✓ Describe the pathogenesis of various types of skin and soft tissue infections
- ✓ Describe the major approaches to treatment of skin and soft tissue infections

Structure of the Skin

- **Skin:**
 - Is a marvelous, flexible, tough membrane that is also called the cutaneous membrane
- **Functions of the Skin:**
 - Prevents excessive water loss
 - Regulates temperature
 - Involved in sensory phenomena
 - Barrier against microbial invaders (when intact)
- **Composed of two main layers:**
 - Epidermis
 - Dermis

Figure 19.1 The skin-overview



Structure of the Skin

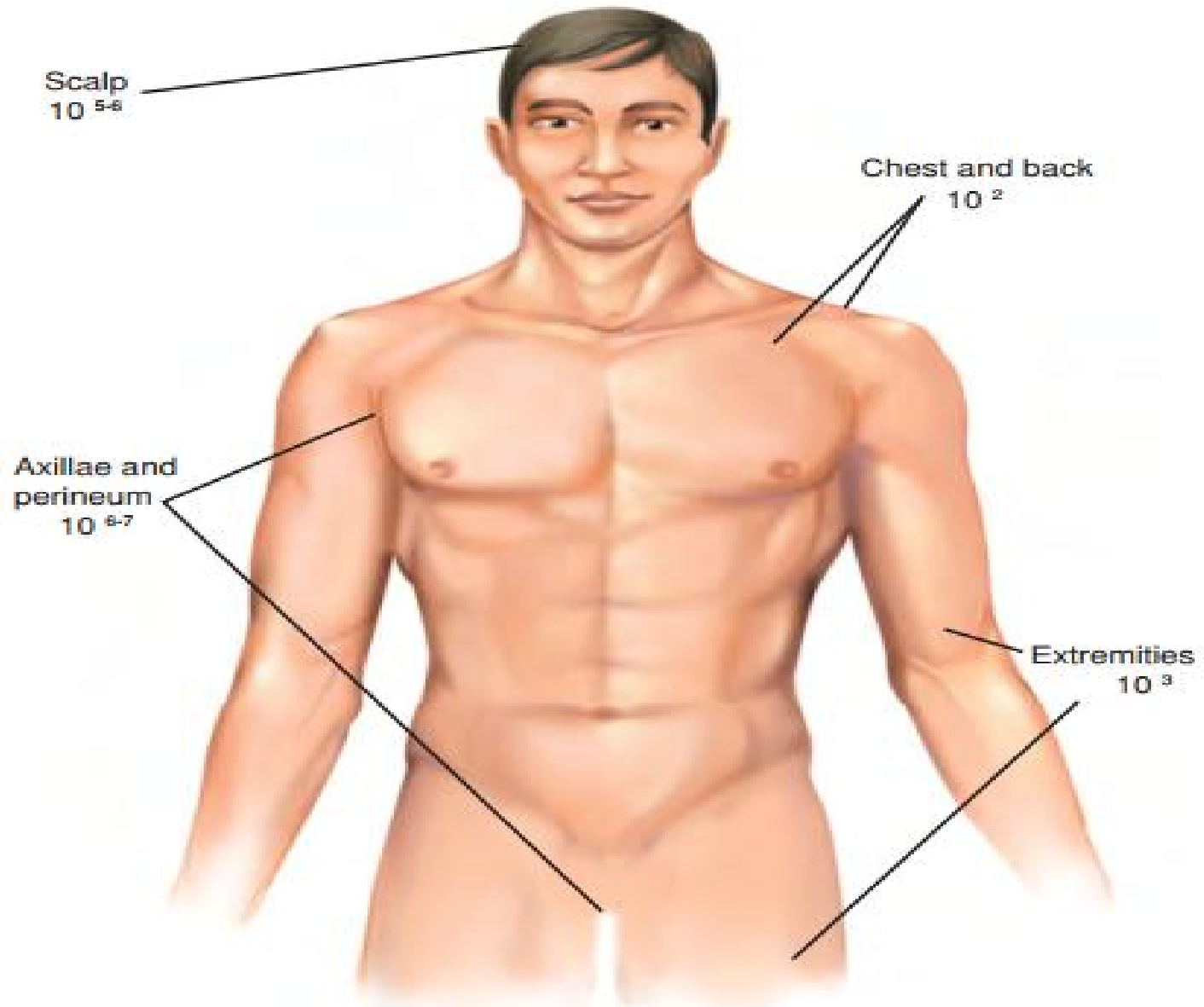
- **Wounds**
 - Trauma to any tissue of the body
 - **Cuts, scrapes, surgery, burns, bites, etc.**
 - Allow microbes to infect the deeper tissues of the body
 - In most cases other body defenses eliminate infection
 - Can result in severe or fatal diseases

Normal Microbiota of the Skin

- **Skin microbiota are normally harmless microbes present on the skin**
 - Cannot be completely removed through cleansing
 - Made up of various microbes
 - **Yeast**
 - *Malassezia*
 - **Bacteria**
 - *Staphylococcus*, *Micrococcus*, and the diphtheroids
 - May produce disease
 - If penetrate epidermis or if immune system is suppressed

Normal Microbiota of the Skin

- ❖ The normal flora (microbes) have the ability to prevent most pathogens:
 - ✓ from colonizing by preventing their attachment to the surface of the skin or
 - ✓ by producing toxins that inhibit the growth of other microorganisms.
- ❖ Some areas of the skin are colonized more heavily than others



Bacterial Diseases of the Skin

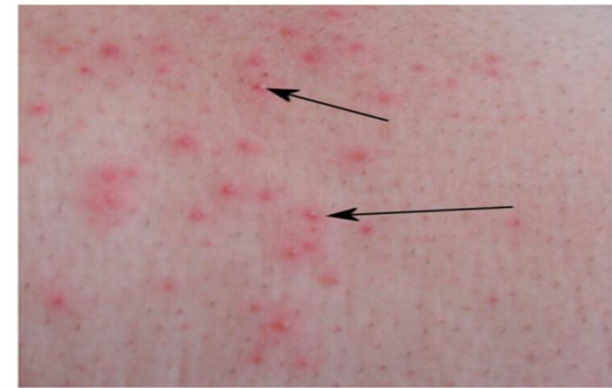


Figure 4-13. The inner thigh of a person with folliculitis caused by shaving the legs.

- **Folliculitis**

- Infection of the hair follicle in which the base of the follicle becomes red, swollen and pus filled
- Often called a **pimple**
- Called a **sty** when it occurs at the eyelid base
- Spread of infection into surrounding tissues can produce **furuncles**
- **Carbuncles** occur when multiple furuncles grow together

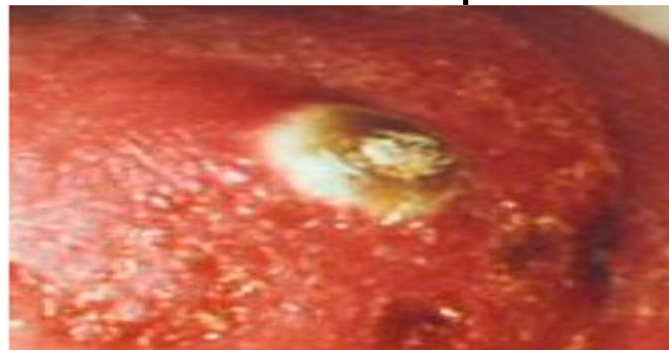


Figure 4-14. A furuncle. Image courtesy of the Centers for Disease Control and Prevention, Joe Miller.



Clinical Manifestations

- Skin
 - folliculitis
 - impetigo (bullous & pustular)
 - scalded skin syndrome
- Primary staphylococcal pneumonia
- Food poisoning vs. foodborne disease
 - Toxic shock syndrome



Clinical Manifestations.....

Impetigo



Folliculitis



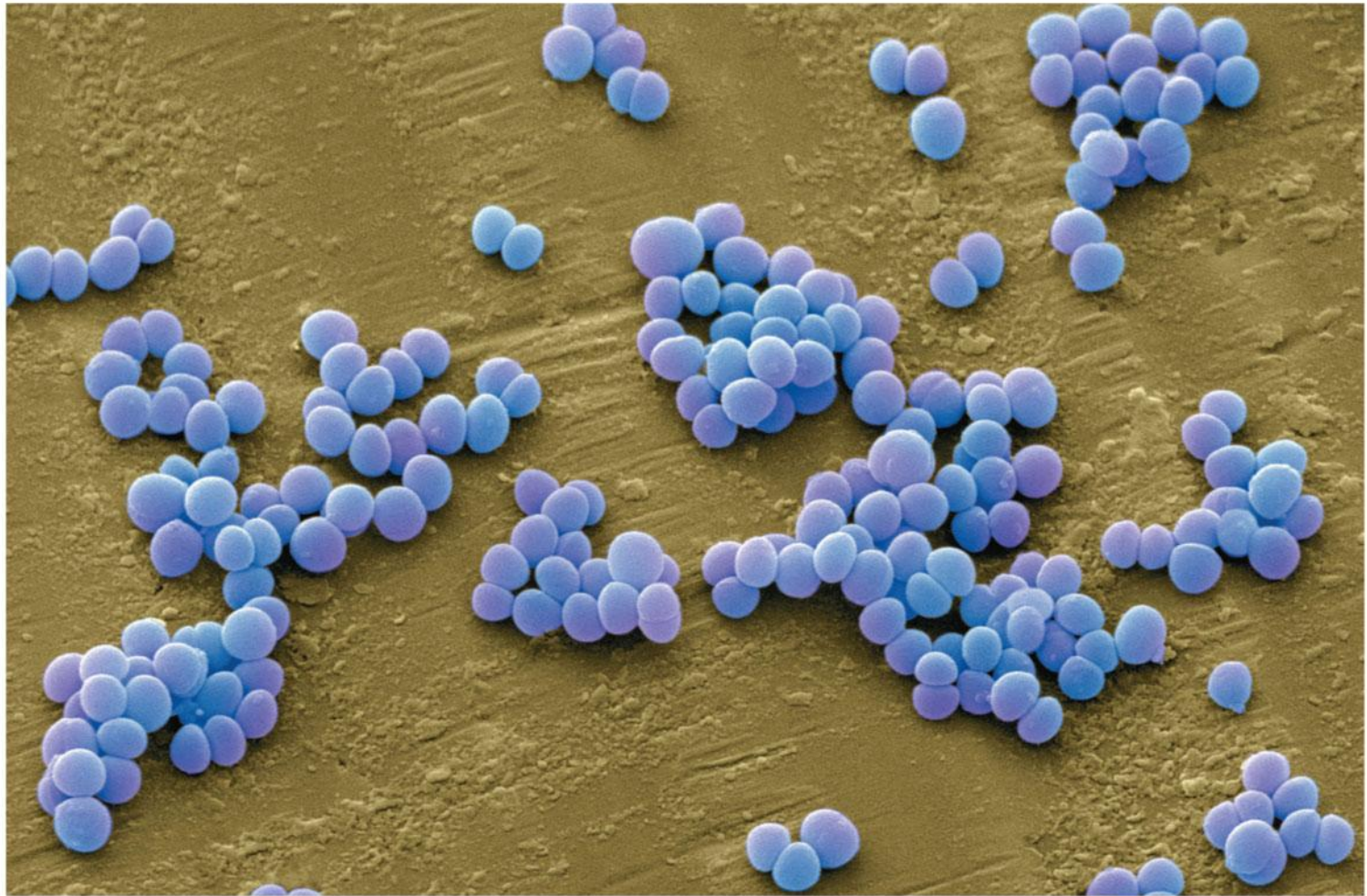
carbuncle

Bacterial Diseases of the Skin

Pathogen and virulence factors

- Most commonly caused by *Staphylococcus*
 - Facultatively anaerobic, Gram-positive bacteria
 - » Cocci typically arranged in clusters
 - **Tolerant of salt**, desiccation, solar radiation & heat
- Two species commonly found on the skin
 - *Staphylococcus epidermidis* (*Dominant microbiota on skin =90%*)
 - *Staphylococcus aureus* (*Grow in nasal passages & more virulent*)
- Staphylococci have 3 categories of virulence factors.
Enzymes, toxins & structures

Figure 19.2 *Staphylococcus*



SEM

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Table 19.1 Comparison of Virulence Factors of Two Staphylococcal Species

19.1

TABLE

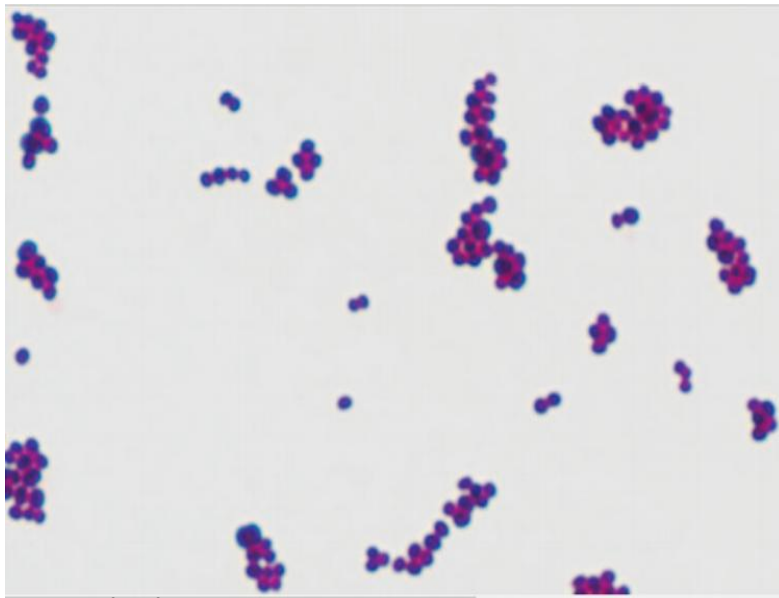
Comparison of Virulence Factors of Two Staphylococcal Species

Virulence Factor	<i>S. aureus</i>	<i>S. epidermidis</i>
Enzymes		
Coagulase	+	—
Staphylokinase	+	—
Lipase	+	+
β -Lactamase	Present in 90% of strains	—
Factors That Inhibit Phagocytosis		
Polysaccharide slime layer	+	+
Protein A on cell surface	+	—
Toxins		
Cytolytic toxins	+	—
Leukocidin	+	—
Epidermal cell differentiation inhibitor	+	—
Exfoliative toxin	Present in some strains	—
Toxic shock syndrome toxin	Present in some strains	—

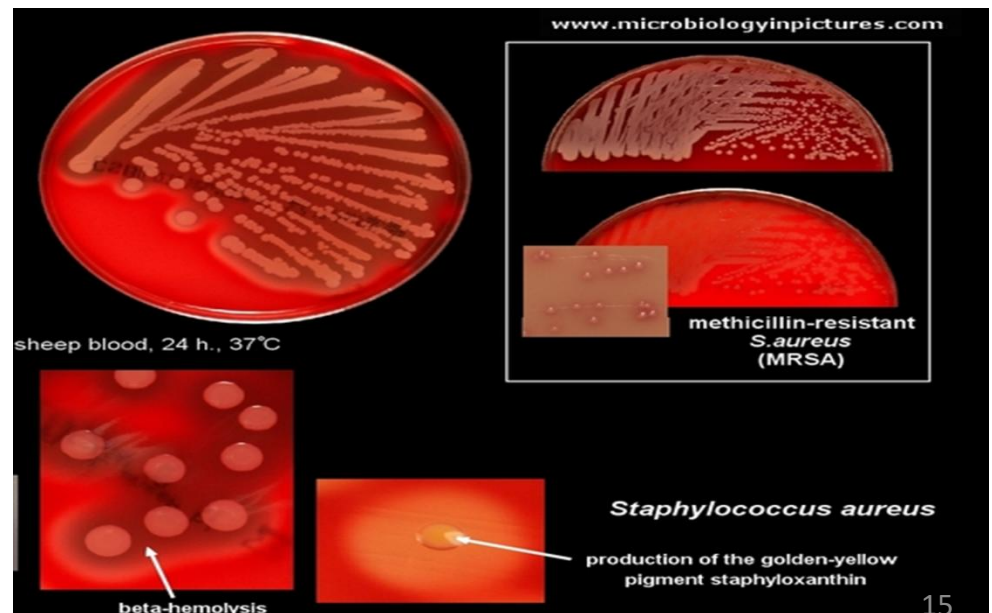
Bacterial Diseases of the Skin

□ Diagnosis

- Diagnosis is usually determined based on clinical appearance of the skin lesions.
- Definitive diagnosis can be obtained by
 - Isolation of Gram-positive bacteria in grapelike clusters from pus
 - Culturing the bacteria obtained from the pus



9/24/2026



Bacterial Diseases of the Skin

– Treatment

- Dicloxacillin (semisynthetic penicillin, not inactivated by β -lactamase) is the drug of choice
- **Vancomycin** used treating strains resistant to penicillin, methicillin, macrolides, aminoglycosides & cephalosporin

– Prevention

- Hand antisepsis
- **Proper procedures in hospitals** to minimize MRSA infections

Bacterial Diseases of the Skin

- **Staphylococcal Scalded Skin Syndrome(SSSS)**
 - Cells of the outer epidermis separate from one another and from the underlying tissue
 - Pathogen and virulence factors
 - Some *Staphylococcus aureus* strains
 - One or two different **exfoliative toxins** cause SSSS
 - Pathogenesis
 - The blood carries **exfoliative toxins** from sites of infection throughout the body
 - **Toxin is a serine protease** that cleaves *desmoglein* in desmosomes
 - **Desmosomes:** are intercellular bridge proteins that hold together cytoplasmic membranes of adjoining cells.

Bacterial Diseases of the Skin

- leading to the separation of the epidermis at the granular cell layer
- No scarring because dermis is unaffected
- The body restores the lost epidermis within 7–10 days
Death is rare but may be due to secondary infections

— Epidemiology

- Disease occurs primarily in infants & children
- Transmitted by person-to-person spread of bacteria onto skin surfaces where it penetrates cuts and abrasions



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▲ **Figure 19.3** Staphylococcal scalded skin syndrome. Exfoliative toxin, produced by some strains of *Staphylococcus aureus*, causes reddened patches of the epidermis to slough off. Which patients are most susceptible to exfoliative strains of *S. aureus*?

Bacterial Diseases of the Skin

- **Staphylococcal Scalded Skin Syndrome**
 - Diagnosis, treatment, and prevention
 - Diagnosed by characteristic sloughing of skin
 - Fluid in the blisters of SSSS does **not contain** *S. aureus*
 - Treated by administration of antimicrobial drugs
 - Widespread presence of *S. aureus* makes prevention difficult



- **Impetigo (Pyoderma)**

- Contagious disease characterized by small, flattened, red patches that appear primarily on the face & limbs
- The patches develop in to oozing, pus-filled vesicles on a red base
- Crusting occurs after the vesicles ruptures
- **Pathogens and virulence factors**
 - Most cases are caused by *S. aureus* (80% of cases)
 - Some cases are caused by *Streptococcus pyogenes*

Bacterial Diseases of



- **Erysipelas**

- Is superficial bacterial skin infection that extends into the cutaneous lymphatic.
- The lesions are raised and well demarcated with shiny bright red areas of dermal inflammation
- **Pathogen and virulence factors**
 - >80% of cases are due to streptococci. *S. pyogenes* is present in over two thirds of cases

- *S. pyogenes*

- Gram-positive coccus, arranged in chains
- Virulence factors similar to those of *S. aureus*
 - » **M protein, Hyaluronic acid capsule, Pyrogenic toxins**

Bacterial Diseases of the Skin

Virulence factors of *Streptococcus pyogenes*

- **M protein:** cytoplasmic membrane component that destabilizes complement and interferes with phagocytosis and lysis of the bacterium.
- **Hyaluronic acid capsule:** serves to “camouflage” the bacterium, hiding it from phagocytes
- **Pyrogenic toxins:** proteins that stimulate macrophages & helper T lymphocytes to release cytokines that in turn stimulate fever, a widespread rash, and shock

Figure 19.4 Impetigo



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▲ **Figure 19.4 Impetigo.** Reddened patches of skin become pus-filled vesicles that eventually crust over.

Figure 19.5 Erysipelas



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Figure 3-18. A patient with erysipelas. Notice the orange peel ("peau d'orange") appearance, the bullae, and the purulent material draining from the eye. Image courtesy of the Centers for Disease Control and Prevention, Dr Thomas F Sellers, Emory University, Atlanta, GA.

Bacterial Diseases of the Skin

- **Impetigo (Pyoderma) and Erysipelas**
 - Pathogenesis
 - The bacteria invade where the skin integrity is compromised
 - Epidemiology
 - Transmitted by person-to-person contact or via fomites
 - **Impetigo** occurs most in children
 - **Erysipelas** can also occur in the elderly

Bacterial Diseases of the Skin

- **Impetigo (Pyoderma) and Erysipelas**
 - Diagnosis, treatment, and prevention
 - The presence of vesicles is diagnostic for impetigo
 - Further, the pus found in the blisters is filled with bacteria and white blood cells
 - Gram stain from the pus and culture
 - **Treat** with topical mupirocin and oral erythromycin/dicloxacillin & careful cleaning of infected areas with soap and water
 - Prevent with proper hygiene and cleanliness

CRITICAL THINKING

Why do impetigo and erysipelas occur more commonly in children than in young adults?

Bacterial Diseases of the Skin

Necrotizing Fasciitis (flesh-eating disease)

- Invasive infection characterized by rapidly progressing inflammation & necrosis of the skin and fascia
- Pathogen and virulence factors
 - Most cases caused by *S. pyogenes*
 - Various enzymes facilitate invasion of tissues
 - **Deoxyribonucleases** (which break down DNA),
 - **Streptokinases** (which dissolve blood clots),
 - **Hyaluronidase** (which breaks down hyaluronic acid between cells)
 - **Exotoxin A** cause toxemia and triggers an overactive response by the immune system, contributing to tissue damage
 - **Streptolysin S** kills many kind of cells

Bacterial Diseases of the Skin

Necrotizing Fasciitis

– Pathogenesis and epidemiology

- *S. pyogenes* enters through breaks in the skin
- Usually spread person-to-person
- Necrotizing fasciitis has developed following surgery and abortion

Risk factors

- Immune-suppression
- **Chronic diseases:** (diabetes, liver and kidney diseases, malignancy)
- **Trauma:**(laceration, cut, abrasion, contusion, burn, bite, subcutaneous injection, operative incision)

Figure 19.6 Necrotizing fasciitis

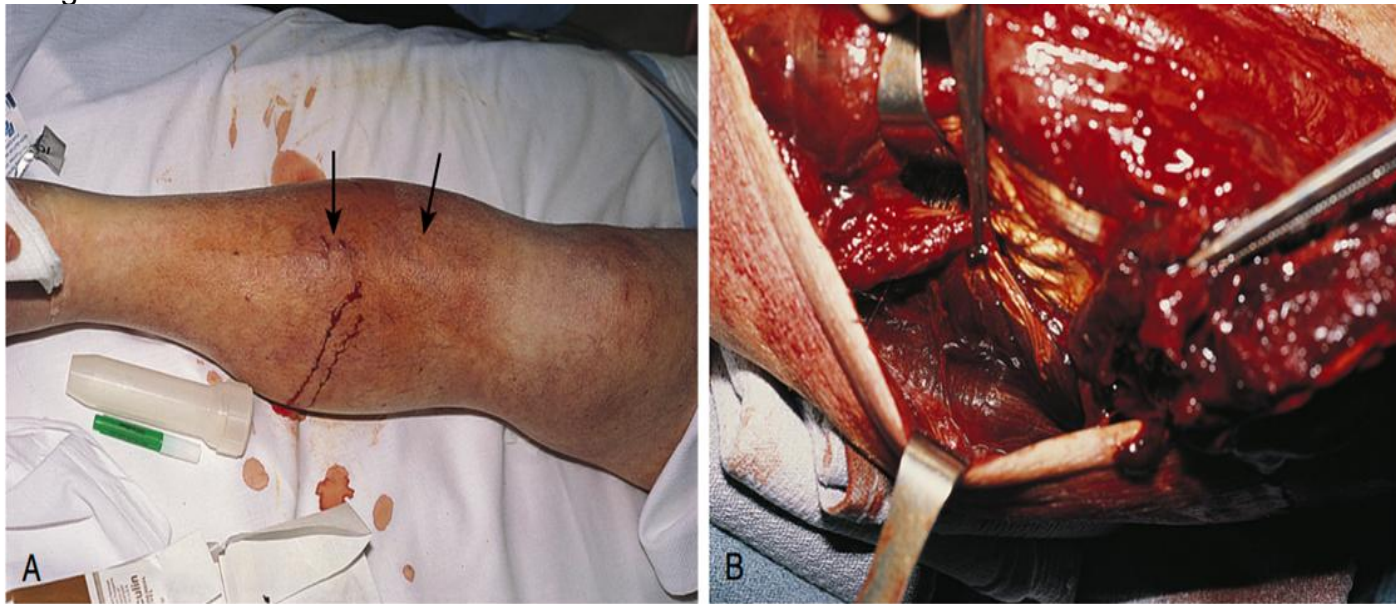


Fig. 19.4 Necrotizing fasciitis caused by *Streptococcus pyogenes*. The patient presented with a 3-day history of malaise, diffuse myalgia, and low-grade fever. Over 3 hours, the pain became excruciating and was localized to the calf. (A) Note the two small, purple bullae over the calf (arrows). (B) Extensive necrotizing fasciitis was present on surgical exploration. The patient died despite aggressive surgical and medical management. (From Cohen, J, Pow-





flesh-eating disease

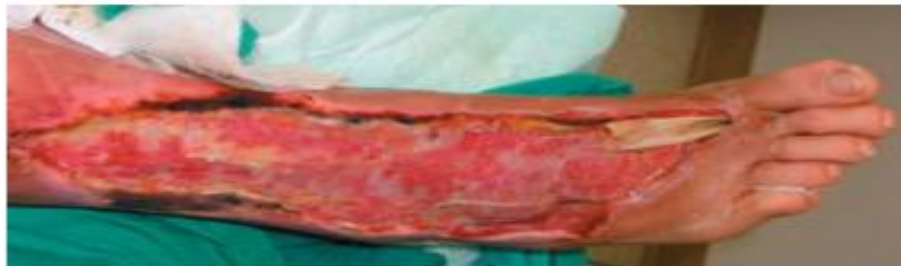
Summary

DISEASE AT A GLANCE 19.1

Necrotizing Fasciitis

Cause: *Streptococcus pyogenes* (group A *Streptococcus*; aerobic, Gram-positive, spherical, bacterium in chains) is the primary cause; other Gram-positive and Gram-negative bacteria, including *Vibrio vulnificus*, *Clostridium perfringens*, *Staphylococcus aureus*, and *Bacteroides fragilis*, may cause the condition.

Portal of entry: Bacteria enter through breaks in the skin, which may be very minor.



▲ Necrotizing fasciitis.

Signs and symptoms:

Intense pain and swelling at site of infection; discoloration and distension of skin, which is hot to the touch; dead (necrotic) tissue; fever, nausea, malaise, and other flulike symptoms; confusion resulting from extremely low blood pressure.

Incubation period: Immediate response; tissue destruction may occur so rapidly as to be visible within an hour.

Susceptibility: Individuals with broken or abraded skin; diabetic, cancer, surgery, and chickenpox patients.

Treatment: Immediate intravenous, broad-spectrum antimicrobial drugs; removal of necrotic tissue.

Prevention: Cleanse wounds thoroughly; consider as extremely dangerous wounds that are severely more tender and painful than it appears they should be, and treat immediately.

Bacterial Diseases of the Skin

Diagnosis, treatment, and prevention

- Early diagnosis is difficult because symptoms are nonspecific
- Affected tissue must be removed completely and aggressively to prevent spread of the bacterium
 - Treat with clindamycin and penicillin
- Difficult to prevent necrotizing fasciitis because *S. pyogenes* is commonly found on humans.

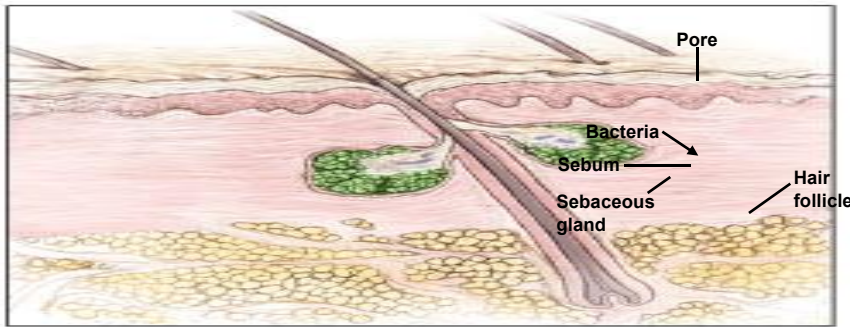
Bacterial Diseases of the Skin

- **Acne**

- Pathogen and virulence factors
 - Commonly caused by ***Propionibacterium acnes***
 - Gram-positive, rod-shaped diphtheroids
 - Commonly found on the skin
- Epidemiology
 - Propionibacteria are normal microbiota
 - Typically begins in adolescence but can occur later in life

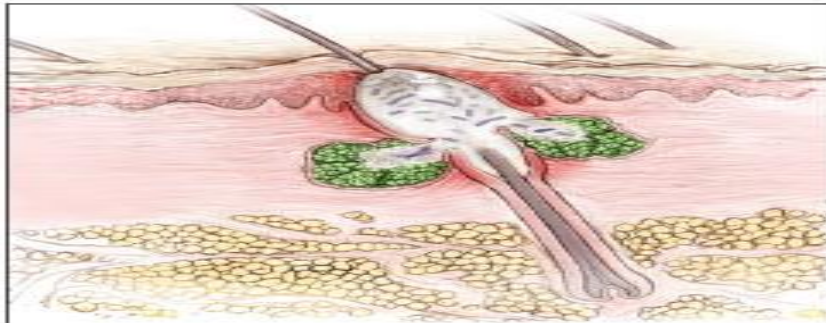


Figure 19.7 The development of acne



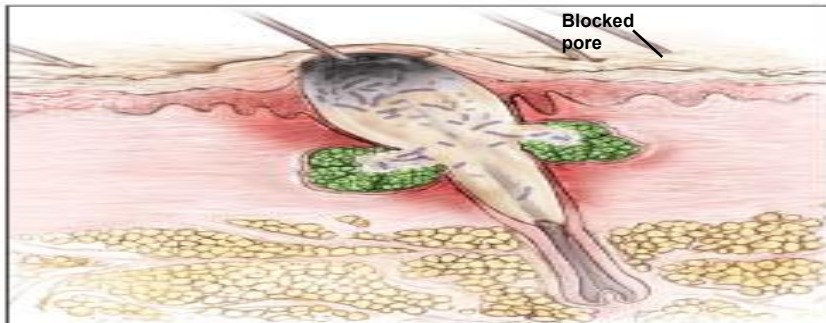
Normal skin

Oily sebum produced by glands reaches the hair follicle and is discharged onto the skin surface via the pore.



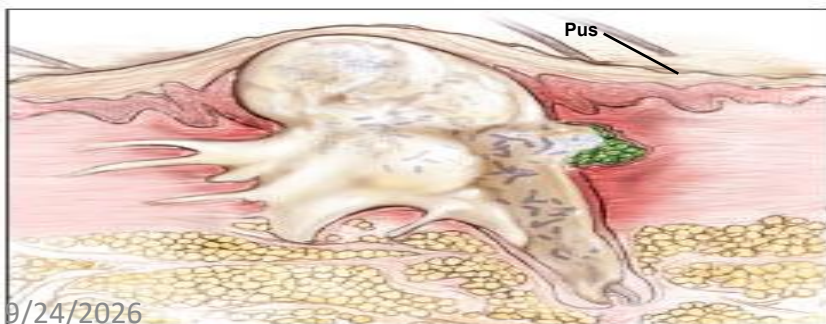
Whitehead

Inflamed skin swells over the pore when bacteria infect the hair follicle, causing the accumulation of colonizing bacteria and sebum.



Blackhead

Dead and dying bacteria and sebum form a blockage of the pore.



Pustule formation

Severe inflammation of the hair follicle causes pustule formation and rupture, producing cystic acne, which is often resolved by scar tissue formation.

Bacterial Diseases of the Skin

- **Acne**
 - Diagnosis, treatment, and prevention
 - Diagnosed by visual examination of the skin
 - Treated with antimicrobial drugs and drugs that cause exfoliation of dead skin cells
 - Accutane is used to treat severe acne
 - New treatment uses blue-light wavelength to destroy bacteria

Bacterial Diseases of the Skin

- ***Pseudomonas* Infection**

- Pathogen and virulence factors

- *Pseudomonas aeruginosa* is the causative agent
 - Gram-negative, aerobic bacillus, Found in soil, decaying matter, moist environments
- Virulence factors
 - **Adhesins** enable its attachment to host cells & formation of biofilms
 - It releases a large amount of potent **endotoxin** upon cell death.
 - **Exotoxin A and exoenzyme S**- Inhibit eukaryotic protein synthesis, contributing to host cell death.
 - **Elastase** breaks down elastic fibers, degrades complement components, and cleaves IgA and IgG
 - **polysaccharide capsule** used for attachment and shields the bacterium from phagocytosis

Bacterial Diseases of the Skin

- ***Pseudomonas* Infection**

- Pathogenesis

- Infection can occur in burn victims
 - Bacteria grow under the surface of the burn
- The bacteria kills cells, destroys tissue, and its endotoxin mediates fever, vasodilation, inflammation and shock

Figure 19.8 *Pseudomonas aeruginosa* infection



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• Pseudomonas Infection Wounds

– Epidemiology

- Natural inhabitant are water bodies and moist soil
- *P. aeruginosa* is rarely part of the microbiota
 - Can cause infections throughout the body once inside

– Diagnosis, treatment, and prevention

- Diagnosis can be difficult because **culture** may represent contamination
 - **Pyocyanin discoloration indicates massive infection**
- Difficult to treat due its resistance to antimicrobial drugs, soaps, antibacterial dyes & quaternary ammonium disinfectants
- Physicians treat infections with a combination of aminoglycoside and beta-lactam antimicrobials

DISEASE AT A GLANCE 19.2

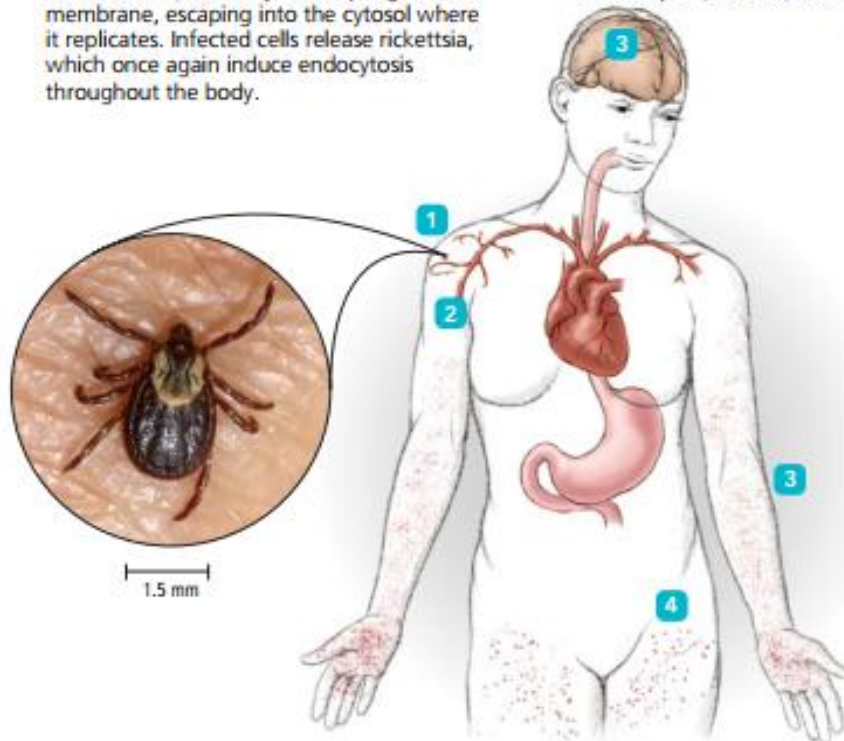
Rocky Mountain Spotted Fever

1 Infected tick (*Dermacentor*) introduces *Rickettsia* while feeding.

2 *Rickettsia* triggers endocytosis by cells lining blood vessels; then it lyses the phagosome's membrane, escaping into the cytosol where it replicates. Infected cells release rickettsia, which once again induce endocytosis throughout the body.

3 Patient develops non-itchy rash on appendages, including palms and soles, that spreads to trunk; other manifestations may include fever, headache, hearing loss, chills, muscle pain, nausea, and vomiting.

4 Damage to cells lining blood vessels allows blood to escape, producing petechiae, low blood pressure, and insufficient delivery of oxygen and nutrients to the body's cells.



The rash in a case of Rocky Mountain spotted fever.

Cause: *Rickettsia rickettsii* (aerobic, Gram-negative, obligate intracellular, rod-shaped bacterium).

Portal of entry: Skin via bite of infected tick.

Signs and symptoms: A non-itchy, spotted rash on the trunk and appendages,

including the palms and soles; and fever, headache, chills, muscle aches, nausea, vomiting, and petechiae.

Incubation period: 5 to 10 days.

Susceptibility: People in highly endemic areas are particularly susceptible due to increased exposure.

Treatment: Doxycycline is the drug of choice, although tetracycline and chloramphenicol are also effective.

Prevention: Wear light-colored, tight-fitting clothes that limit tick exposure. Use tick repellent, promptly remove ticks, examine skin for ticks and bites, and if possible, avoid tick-infested areas.

Cutaneous Anthrax

Pathogen and virulence factors

– Caused by *Bacillus anthracis*

- Gram positive, aerobic, rod shaped, spore former, soil inhabitant

– Virulence factor

- Has protein Capsule

- Protects against phagocytic engulfment and destruction

- Anthrax toxin, Which has three components

1. Protective antigen (PA)

EF+PA= Edema toxin

2. Edema factor (EF)

LF+PA= Lethal toxin

3. Lethal factor (LF)

Bacterial Diseases of the



Cutaneous anthrax

- begins when *Bacillus anthracis*, shed by an infected animal, enters a wound in the skin, producing a solid skin nodule.
- Cells in the affected area die, and the nodule spreads to form a painless, swollen, black, crusty ulcer called an **eschar**

Cutaneous Anthrax

Pathogenesis

- Spores enter the broken skin, germinate and rapidly proliferate at the site of entry
- Characterized by an **eschar**
 - Black, painless, ulcer
- Treated with antimicrobial drugs like ciprofloxacin, penicillin, or erythromycin for 60 days.
- Prevention requires **Vaccination of animals, deep burial or burning of animals, give vaccination for people**



Cutaneous anthrax skin infection



ADAM.

DISEASE AT A GLANCE 19.3

Cutaneous Anthrax

Cause: *Bacillus anthracis* (facultatively anaerobic, endospore-forming, rod-shaped bacterium).

Portal of entry: Direct contact of endospores with wounded skin, infected animals, or infected animal products. (Two other forms of anthrax—gastrointestinal and inhalation—have routes of entry via the gastrointestinal tract and respiratory tract, respectively. More on anthrax is found in Chapter 22.)



▲ Eschars.

Signs and symptoms: Localized itching followed by a raised lesion which eventually forms a painless, black eschar (see photo) within 7–10 days.

Incubation period: Immediate response, within one day.

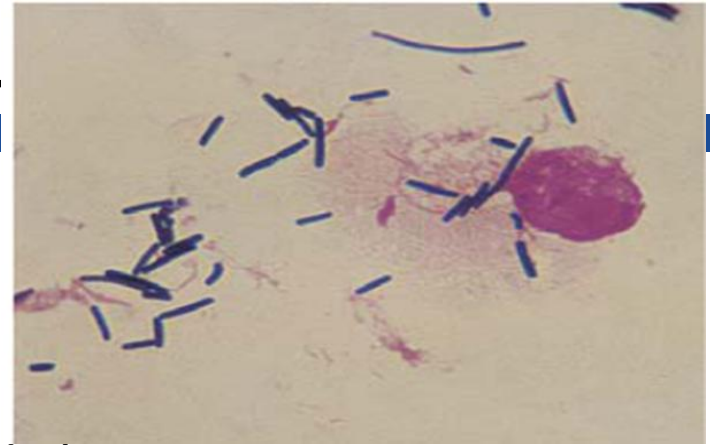
Susceptibility: The most commonly infected individuals are animal handlers.

Treatment: Ciprofloxacin is the preferred treatment, but penicillin or erythromycin are also effective. There is no treatment for the toxin.

Prevention: Vaccinate livestock. The vaccine, considered safe for humans but recommended only for high-risk personnel, requires six doses over an 18-month period and annual boosters.

Gas Gangrene

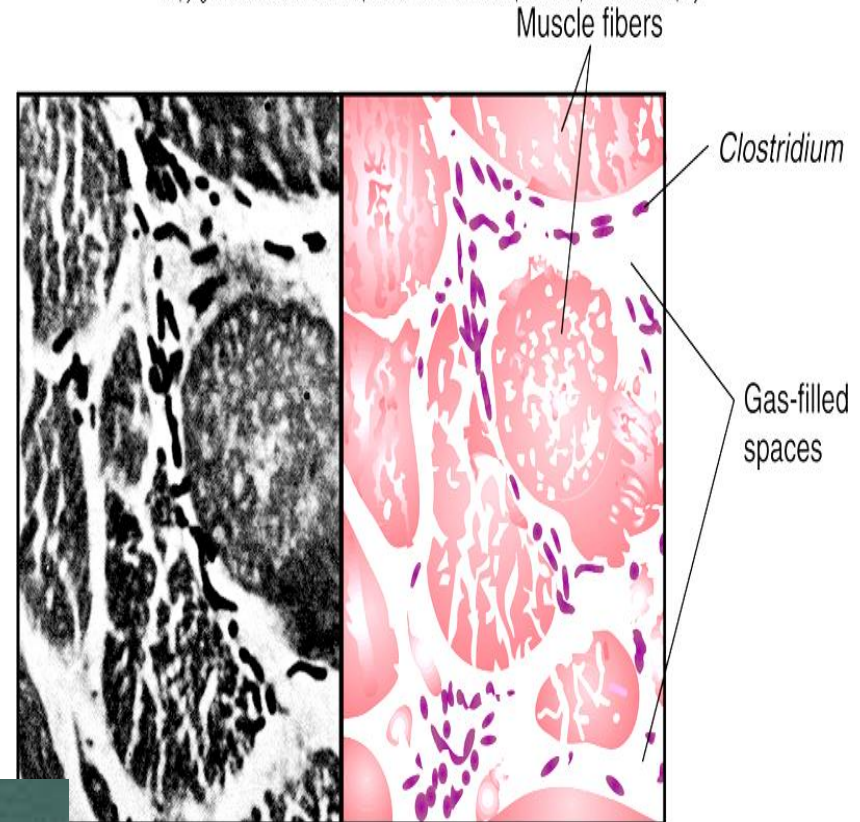
- Signs and symptoms
 - Blackening of infected muscle and skin
 - Presence of gas bubbles
- Pathogens and virulence factors
 - Clostridium is an **anaerobic, Gram-positive rod, non motile, endospore-forming** & ubiquitous in soil, water, sewage & GI of animals and humans
 - Pathogenicity is due ability of
 - Endospores to survive harsh conditions
 - Vegetative cells' secretion of 11 toxins
 - ***C. perfringens***, the species most frequently isolated from patients



Gas Gangrene Pathogenesis

- requires **damaged and dead tissue** and **anaerobic** conditions
- This stimulates **spore germination**, vegetative growth and release of exotoxins, and other virulence factors.
- Fermentation of muscle carbohydrates results in the **formation of gas and further destruction of tissue.**

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Bacterial Diseases of the Skin

Gas Gangrene

– Pathogenesis and epidemiology

- Traumatic event must introduce endospores into dead tissue, crush injuries, gunshot wounds, surgical incision
- Mortality rate exceeds 40%

– Diagnosis, treatment, and prevention

- Appearance is usually diagnostic, though the detection of large Gram-positive bacilli is confirmatory
- Rapid treatment is crucial
 - Surgical removal of dead tissue
 - Administration of antitoxin and penicillin
 - Oxygen applied under high pressure
- Prevent with proper cleaning of wounds, Prophylaxis

DISEASE AT A GLANCE 19.4

Gas Gangrene

Cause: *Clostridium* species, particularly *C. perfringens* (anaerobic, Gram-positive, endospore-forming rod).

Portal of entry: *Clostridium* endospores are introduced into dead tissue via a traumatic event such as surgical incision, puncture, gunshot wound, crushing trauma, abortion, or broken bones punching through the skin.

Signs and symptoms: Increased pain and swelling around injury, fever, initially pale skin that turns dusky and then progresses to dark red or purple, foul-smelling drainage from tissues, crepitation (crackly sound due to gas under the skin), and tachycardia. Untreated gas gangrene causes shock, coma, and death.

Incubation period: Generally less than 24 hours and no more than three days.

Susceptibility: Individuals with deep, lacerating wounds, particularly "dirty" wounds.

Treatment: Gas gangrene is a medical emergency, and prompt identification and treatment is crucial. Dead tissue must be surgically removed and antitoxin and penicillin administered.

Prevention: Proper cleaning of wounds.



▲ Blackened dead tissue and bubbling of bacterial gas wastes in gas gangrene.

Other Specific Skin Infections

Epidemiology	Common Pathgen(s)	Therapy
Cat/Dog Bites	<i>Pasturella multocida</i> ; <i>Capnocytophaga</i>	Amox/clav (Doxy; FQ or SXT + Clinda)
Human bites	Mixed flora <i>eikenella corrodens</i>	Hand Surgeon; ATB as above
Fresh water injury	<i>Aeromonas</i>	FQ; Broad Spectrum Beta-lactam
Salt water injury (warm)	<i>Vibrio vulnificus</i>	FQ; Ceftazidime
Thorn , Moss	<i>sporothrix schenckii</i>	Potassium iodine
Meat-packing	<i>Erysipelothrix</i>	Penicillin
Cotton sorters	Anthrax	Penicillin
Cat scratch	<i>Bartonella</i>	Azithromycin

Viral Diseases of the Skin

viral skin infections

- Herpes simplex
 - Herpes zoster
 - Warts and molluscum
- Many viral diseases are systemic in nature and spread by respiratory and oral routes
 - These diseases can result in signs and symptoms in the skin
 - These include diseases of:
 - *poxviruses, herpes infections, warts, chickenpox, shingles, rubella, measles, roseola, erythema infectiosum, and coxsackie viral infections*

Viral Diseases of the Skin

- **Diseases of Poxviruses**
 - Poxviruses that cause human diseases
 - Smallpox
 - Orf, cowpox, and monkeypox (rare)
 - Smallpox first human disease eradicated
 - Signs and symptoms
 - Diseases progress through a series of stages

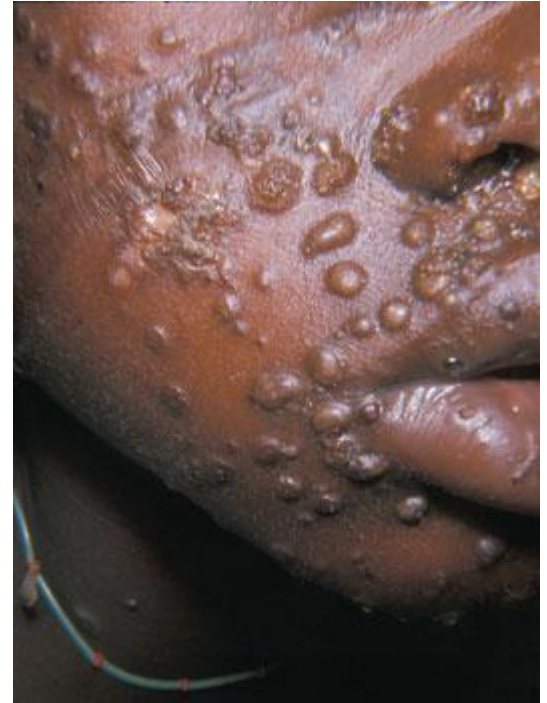
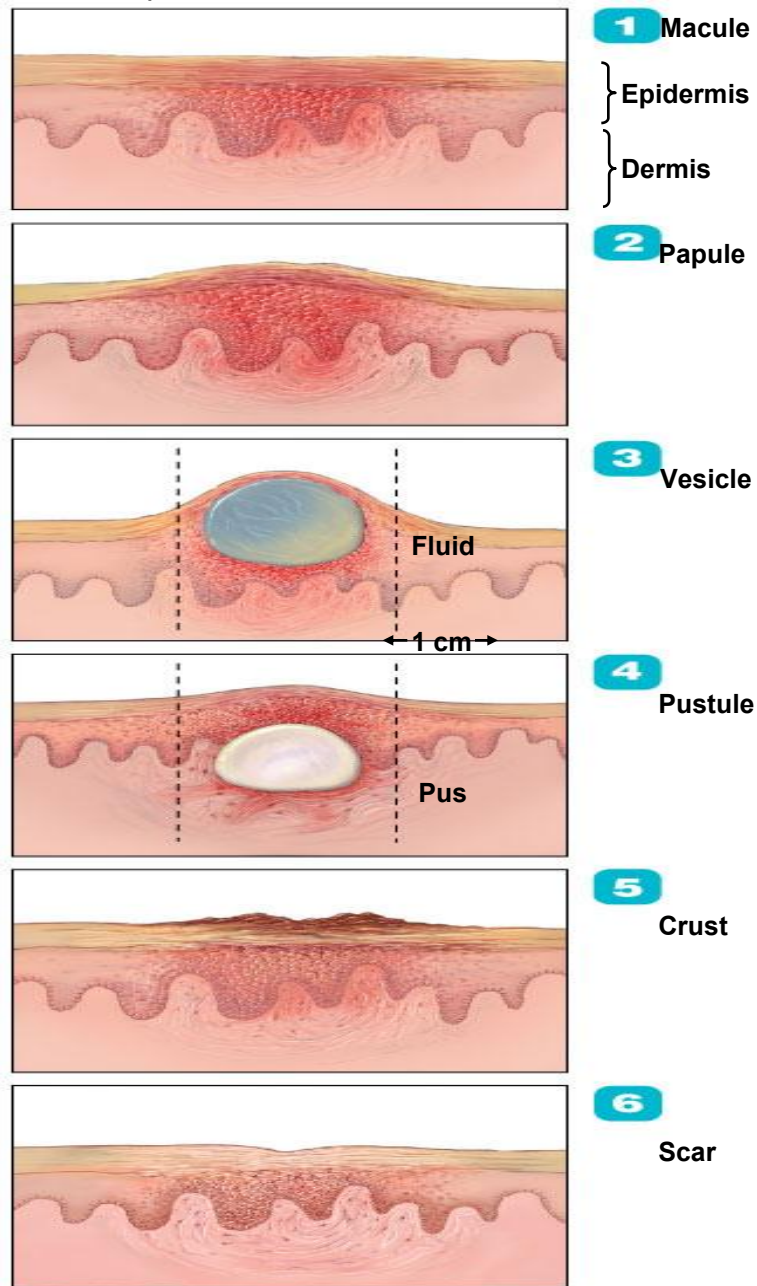


Figure 19.10 The stages of lesions in poxviral skin infections-overview



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Viral Diseases of the Skin

- **Diseases of Poxviruses**

- Pathogens and virulence factors
 - Caused by *Orthopoxvirus* (variola virus)
- Pathogenesis
 - Infection occurs by inhalation of virus
- Epidemiology
 - Increase in monkeypox cases over the past decade
- Diagnosis, treatment, and prevention
 - Treatment requires immediate vaccination
 - Vaccination discontinued in 1980s

CRITICAL THINKING

The United States and Russia have repeatedly agreed to destroy their stocks of the smallpox virus, but the deadline for destruction has been postponed numerous times. In the meantime, the entire genome of variola major has been sequenced. What reasons can governments cite for maintaining smallpox viruses? Should all laboratory stores of smallpox viruses be destroyed? Given that the genome of the virus has been sequenced, and that DNA can be reconstructed if the sequence of nucleotides is known, would elimination of all laboratory stocks really be the extinction of the smallpox virus?

DISEASE AT A GLANCE 19.5

Smallpox

Cause: *Orthopoxvirus* (variola virus) (enveloped, double-stranded DNA virus).

Portal of entry: Inhalation of viruses in droplets or dried crusts.

Signs and symptoms: The prodromal period (two to four days) includes high fever, head and body aches, and malaise.

A rash begins in the mouth and spreads to the face and downward all over the body. The rash becomes papules, then vesicles, and finally pustules. Eventually the pustules dry to form crusts. Scarring may result.

Incubation period: Average 12 to 14 days, but can range from 7 to 17 days. The person is not contagious at this stage.

Susceptibility: Currently, there is no naturally occurring smallpox. However, if a bioterrorism event or accidental release were to occur, it is theorized that most people would be susceptible since regular vaccination was discontinued decades ago.

Treatment: Immediate vaccination; there is no other approved antiviral treatment for smallpox. Treatment includes supportive therapy.

Prevention: Routine vaccination was discontinued in the U.S. in 1972. In response to recent world events, the U.S. government ordered new production of the smallpox vaccine and is currently prepared in the event of an outbreak.



- **Herpes Infections**

- Signs and symptoms

- Slow spreading skin lesions
 - Recurrence of lesions is common

- Pathogen and virulence factors

- Caused by human herpesviruses 1 and 2
 - Produce various proteins that act as virulence factors

- Pathogenesis

- Painful lesions caused by inflammation and cell death
 - Cause fusion of cells to form syncytia

Figure 19.11 Oral herpes lesions



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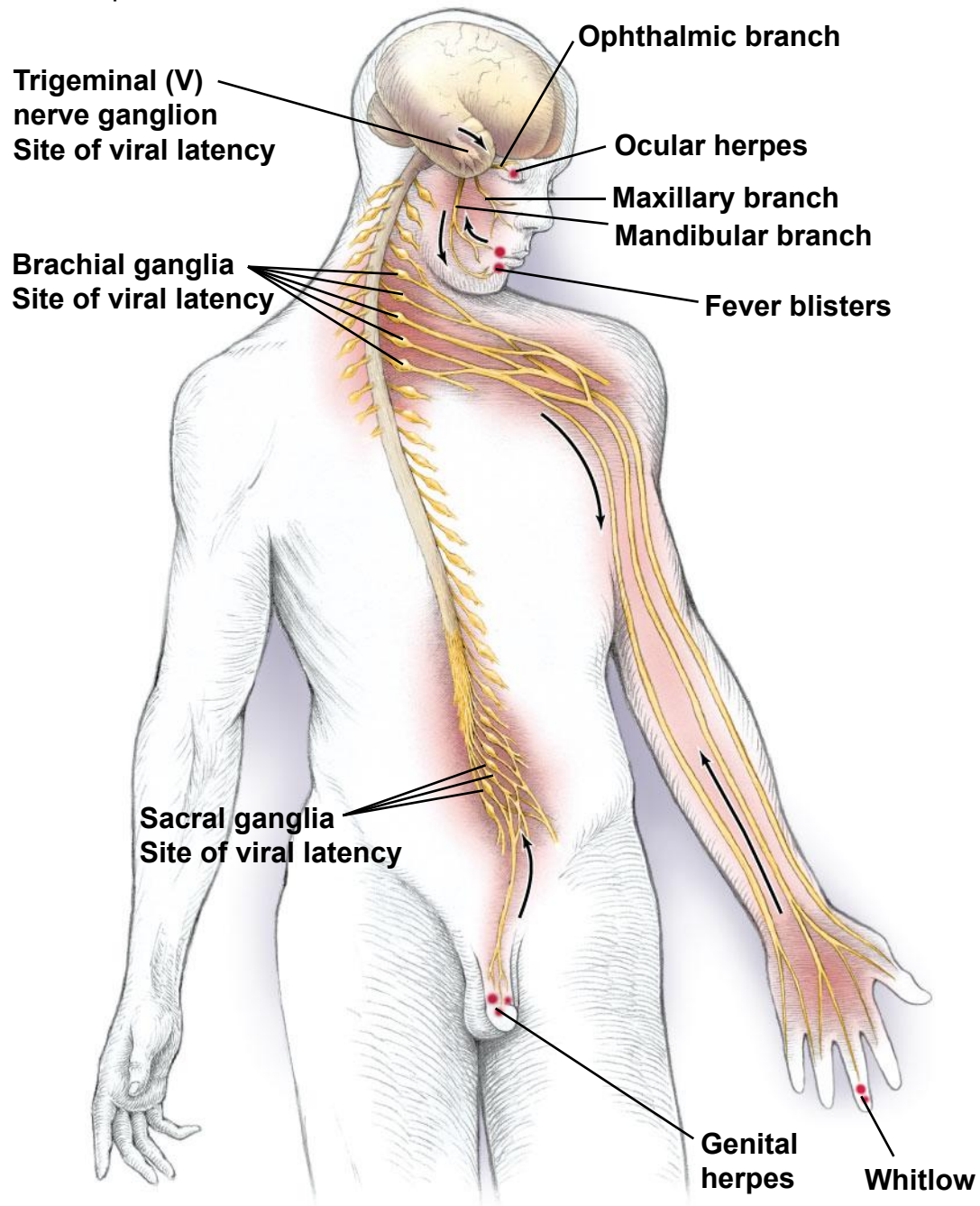
Genital herpes

Most common viral venereal disease in Ethiopia

- any infected person may shed virus asymptomatically.
- **The primary infection is usually more serious, painful genital ulcers, dysuria, fever, tender local lymphadenopathy, and headache.**
- Itchy Painful lesions can be developed
- Shaft of the penis in men (Rarely in urethra)
- Vulva, vagina, cervix, peri-anal region, inner thigh of women associated with itching & Vaginal discharge
- **Homosexuals- Rectal/anal lesions (proctitis)**
- Recurrent genital HSV disease (2 to nine times/year) is shorter in duration (2 to 3 weeks) and less severe than the primary episode
- 50% of patients, recurrences are preceded by a characteristic prodrome of burning or tingling in the area in which the lesions eventually erupt



Figure 19.12 Sites of events in herpesvirus infections



Viral Diseases of the Skin

- **Herpes Infections**

- Epidemiology

- Spread between mucous membranes of mouth and genitals
 - Herpes infections in adults are not life threatening

- Diagnosis, treatment, and prevention

- Diagnosis made by presence of characteristic lesions
 - Immunoassay reveals presence of viral antigens
 - Chemotherapeutic drugs help control the disease but do not cure it

Comparative Epidemiology and Pathology of Human Herpesvirus Infections

	HHV-1	HHV-2
Usual diseases	90% of cold sores/fever blisters; whitlow	85% of genital herpes cases
Mode of transmission	Close contact	Sexual intercourse
Site of latency	Trigeminal and brachial ganglia	Sacral ganglia
Locations of lesions	Face, mouth, and rarely trunk	External genitalia, and less commonly thighs, buttocks, and anus
Other complications	15% of genital herpes cases; pharyngitis; gingivostomatitis, ocular/ophthalmic herpes; herpes gladiatorum; 30% of neonatal herpes cases	10% of oral herpes cases; 70% of neonatal herpes cases

DISEASE AT A GLANCE 19.6

Herpes

Cause: *Human herpesviruses 1 and 2* (enveloped double-stranded DNA viruses).

Portal of entry:

Usually mucous membranes of the mouth or genitals.

Signs and symptoms:

A prodromal period of malaise, fever, and/or muscle pain may precede the typical blisters of a herpes infection. The blisters break, leaving a tender ulcer that may take several weeks to heal.



▲ **Whitlow.**

Incubation period: 10 to 14 days.

Susceptibility: Sexually active people have a greater risk of exposure to the disease, as does a baby during delivery by an infected mother.

Treatment: Topical or oral administration of antivirals such as acyclovir, valacyclovir, or famciclovir can shorten outbreaks, but there is currently no cure.

Prevention: Sexual abstinence, mutual monogamy, use of condoms, use of latex gloves by health care workers, and cesarean sections in infected mothers.

Viral Diseases of the Skin

- **Warts**

- Benign epithelial growths on the skin or mucous membranes
 - Can form on many body surfaces
- Various **papillomavirus strain, like 42, 43, 44, 54 and 61** cause warts on skin or genital area.
- Most warts are harmless
- Transmitted via direct contact and fomites
- Diagnosed by observation
- Various techniques to remove warts
 - New warts can develop due to latent viruses



(a)



(b)

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▲ Figure 19.13 Various kinds of warts—lesions caused by papillomaviruses. (a) Plantar warts on the sole of a foot. (b) Flat warts, which often occur on the trunk, face, elbows, or knees. Genital warts can occur on the genitalia or anal perineum in both sexes.



Figure 3-29. A patient with severe anogenital wart infection. Image courtesy of the Centers for Disease Control and Prevention, Dr Wiesner.



Figure 3-30. Anogenital warts due to human papillomavirus infection on and around the patient's labia. Image courtesy of the Centers for Disease Control and Prevention, Joe Miller.



Fig. 1.18. Condylomata acuminata: In the genital region of a



Fig. 1.17. Condylomata acuminata: Multiple lesions in a male patient, West Africa

Viral Diseases of the Skin

- **Chickenpox and Shingles**

- Signs and symptoms

- Chickenpox characterized by lesions on the back and trunk that spread across body
 - Shingles lesions localized to skin along an infected nerve

- Pathogen

- Varicella-zoster virus (VZV) causes both diseases

- Pathogenesis

- Infected dermal cells cause rash characteristic of chickenpox
 - Virus becomes latent in nerve ganglia
 - Reactivated VZV causes shingles

Viral Diseases of the Skin

- **Chickenpox and Shingles**

- Epidemiology

- Chickenpox occurs mostly in children
 - Disease is more severe in adults
 - Risk of shingles increases with age

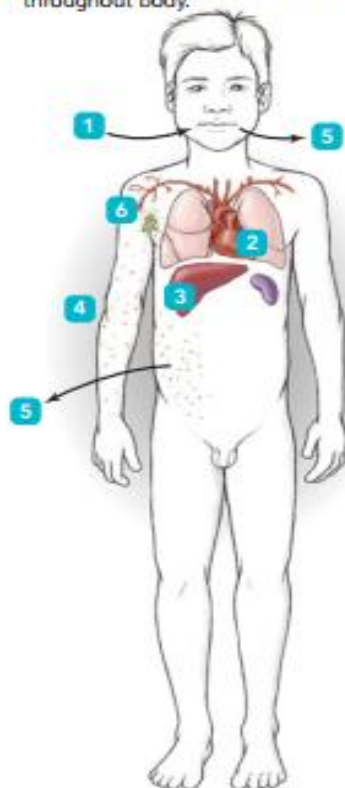
- Diagnosis, treatment, and prevention

- Diagnosis based on characteristic lesions
 - Treatment based on relief of symptoms
 - Vaccine available against chickenpox



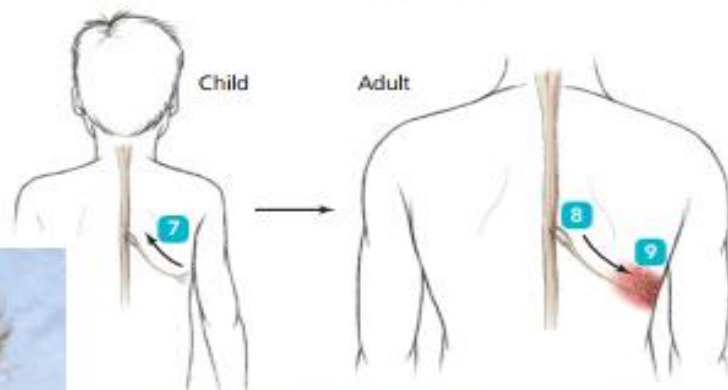
Chickenpox and Shingles

- 1 Inhalation of varicella-zoster virus.
- 2 Infected cells replicate and release virus into the blood (viremia) and lymph.
- 3 Viruses infect liver, spleen, and lymph nodes throughout body.



- 4 Second viremia spreads viruses to skin, causing extensive rash.
- 5 Viruses shed through respiratory droplets and fluid from skin lesions.
- 6 Immune system eliminates viruses from blood and most cells.

- 7 Some viruses become latent (dormant) in nerve ganglia.
- 8 Upon reactivation, viruses travel inside nerve cells down the nerve to infect skin cells.
- 9 Rash develops in the band of skin innervated by the infected nerve.



Characteristic chickenpox lesions



Characteristic shingles rash

Cause: Varicellovirus (varicella-zoster virus, VZV) (enveloped, double-stranded DNA virus).

Portal of entry: Respiratory tract.

Signs and symptoms: Fever, characteristic rash on back and trunk that spreads to face and extremities. In 15–20% of cases, the virus reactivates years later, causing a painful shingles

rash on a band of skin innervated by an infected nerve.

Incubation period: Two to three weeks.

Susceptibility: Chickenpox is a highly contagious disease to those who are not vaccinated or have not had it before. Persons who have had chickenpox previously are generally immune.

Treatment: Treatment is supportive therapy. Symptoms may be treated with acetaminophen and/or antihistamines. Aspirin should never be given to children with viral infections.

Prevention: The CDC recommends the attenuated vaccine for children and a more potent live vaccine for adults 19–49 who had chickenpox as children.

Viral Diseases of the Skin

- **Rubella**

- Children develop a mild rash
- Congenital infection can result in birth defects or death of fetus
- Caused by rubella virus
 - Spread by respiratory secretions
 - Infects only humans
- Diagnosis made by observation of rash and serological testing
- Vaccine is available
 - Aimed at preventing rubella infections in pregnant women

DISEASE AT A GLANCE 19.8

Rubella

Cause: *Rubivirus* (rubella virus) (enveloped, single-stranded RNA virus).

Portal of entry: Respiratory tract.

Signs and symptoms: Mild rash of pink to red macules and swollen lymph nodes in children. Causes severe congenital defects in unborn babies, including deafness, blindness, microcephaly, and mental and growth retardation.

Incubation period: 7 to 14 days.

Susceptibility: Unvaccinated individuals and unborn babies.

Treatment: There is no specific treatment for measles other than supportive therapy.

Prevention: Immunization with the combined MMR (measles, mumps, and rubella) vaccine. Pregnant women should not be given the live, attenuated vaccine.



Disease at a Glance 19.8 summarizes the features of rubella

Viral Diseases of the Skin

- **Measles (Rubeola)**

- Signs and symptoms

- Characterized by Koplik's spots
 - Subacute sclerosing panencephalitis is rare complication

- Pathogen and virulence factors

- Caused by measles virus
 - Adhesion and fusion proteins help virus avoid immune recognition

- Pathogenesis

- Immune response to infected cells causes most symptoms

- **Measles (Rubeola)**
 - Epidemiology
 - Measles is highly contagious
 - Spread via respiratory droplets
 - Humans are the only host
 - Diagnosis, treatment, and prevention
 - Diagnosis based on signs of measles
 - No treatment is available
 - Measles immunization available as part of MMR vaccine

Viral Diseases of the Skin

- **Other Viral Rashes**

- Erythema infectiosum

- Caused by an erythrovirus of family Parvoviridae
 - Respiratory disease that manifests as a rash
 - Also referred to as fifth disease

- Roseola

- Caused by human herpesvirus 6 (HHV-6)
 - Characterized by a rose-colored rash

- Coxsackievirus infection

- Caused by coxsackie A viruses
 - Produces lesions like those from herpes infections
 - Also causes hand-foot-and-mouth disease

Figure 19.16 A case of erythema infectiosum (fifth disease)



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DISEASE AT A GLANCE 19.9

Measles

Cause: *Morbillivirus* (rubeola virus) (enveloped single-stranded RNA virus).

Portal of entry: Respiratory tract.

Signs and symptoms: Fever, sore throat, headache, dry cough, conjunctivitis, Koplik's spots, and maculopapular rash.

Incubation period: 8 to 12 days.

Susceptibility: Unvaccinated individuals.

Treatment: There is no specific treatment for measles other than supportive therapy.

Prevention: Immunization with the combined MMR (measles, mumps, and rubella) vaccine. The vaccine is a live, attenuated vaccine.

Koplik's spots in a mouth





Measles



Rubella

A Comparison of Measles and Rubella

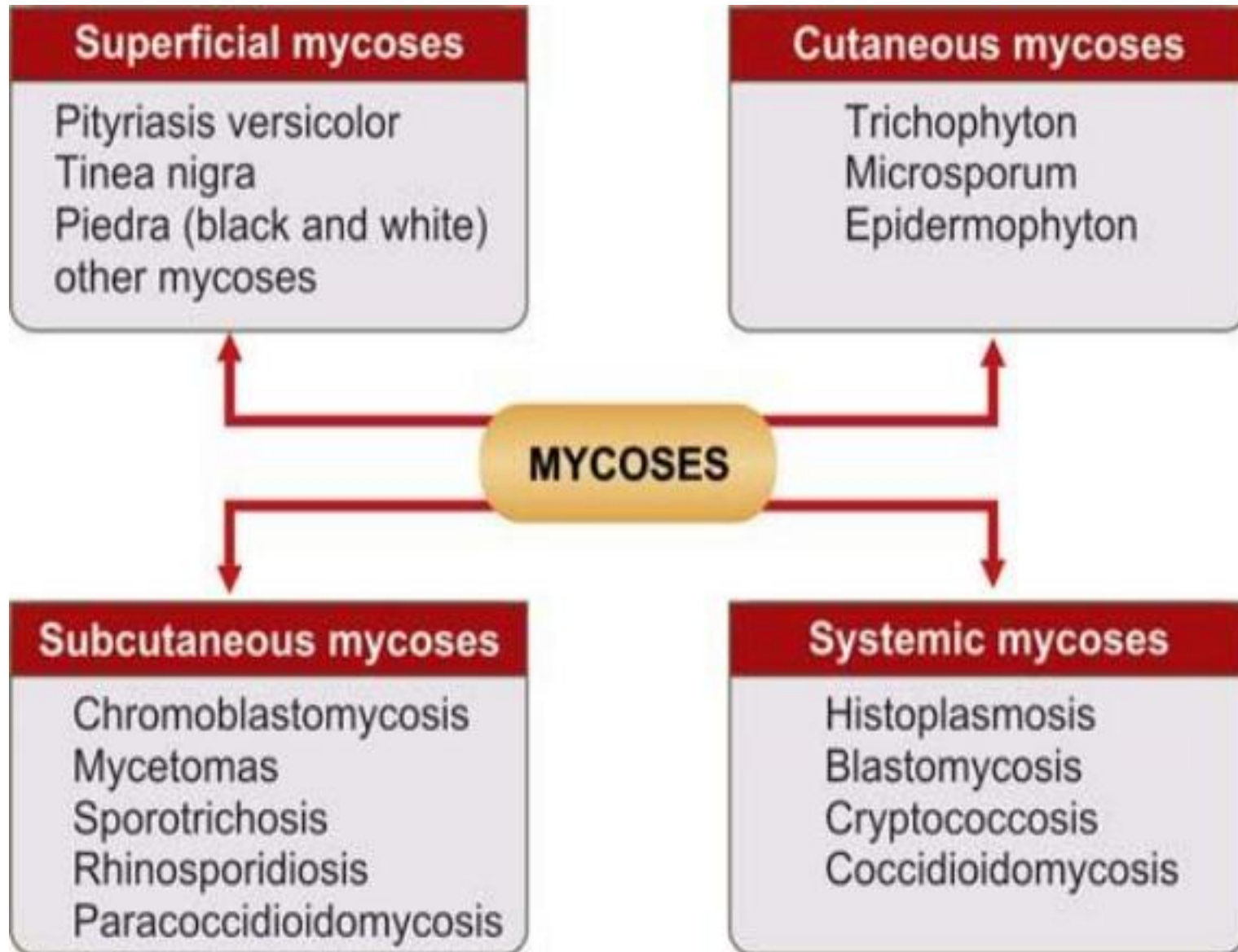
Disease	Causative Agent	Primary Patient(s)	Complications	Skin Rash	Koplik's Spots
Rubella (also known as German measles, rubeola, or three-day measles)	<i>Togaviridae:</i> <i>Rubivirus</i>	Child, fetus	Birth defects	Mild	Absent
Measles (also known as rubeola or red measles)	<i>Paramyxoviridae:</i> <i>Morbillivirus</i>	Child	Pneumonia, encephalitis, subacute sclerosing panencephalitis	Extensive	Present

SKIN MYCOSES

(Fungal infection of the skin)

Mycoses of the Hair, Nails, and Skin

- Mycoses are diseases caused by fungi
 - Most are opportunistic pathogens
 - Mycoses are classified by infection location
 - **skin mycoses**
 - superficial mycoses
 - cutaneous mycoses
 - subcutaneous mycoses
 - **systemic mycoses**
 - systemic mycoses due to primary (usually dimorphic) pathogens
 - systemic mycoses due to opportunistic pathogens.
-
- **Superficial** – occur on the **hair**, **nails**, and outer **skin** layers; most common fungal infections
 - **Cutaneous**– Infections that extend deeper into the epidermis as well as invasive hair and nail disease
 - **Subcutaneous** – in the hypodermis and muscles
 - **Systemic** – affect numerous systems



Mycoses of the Hair Nails And Skin

1. Superficial Mycoses

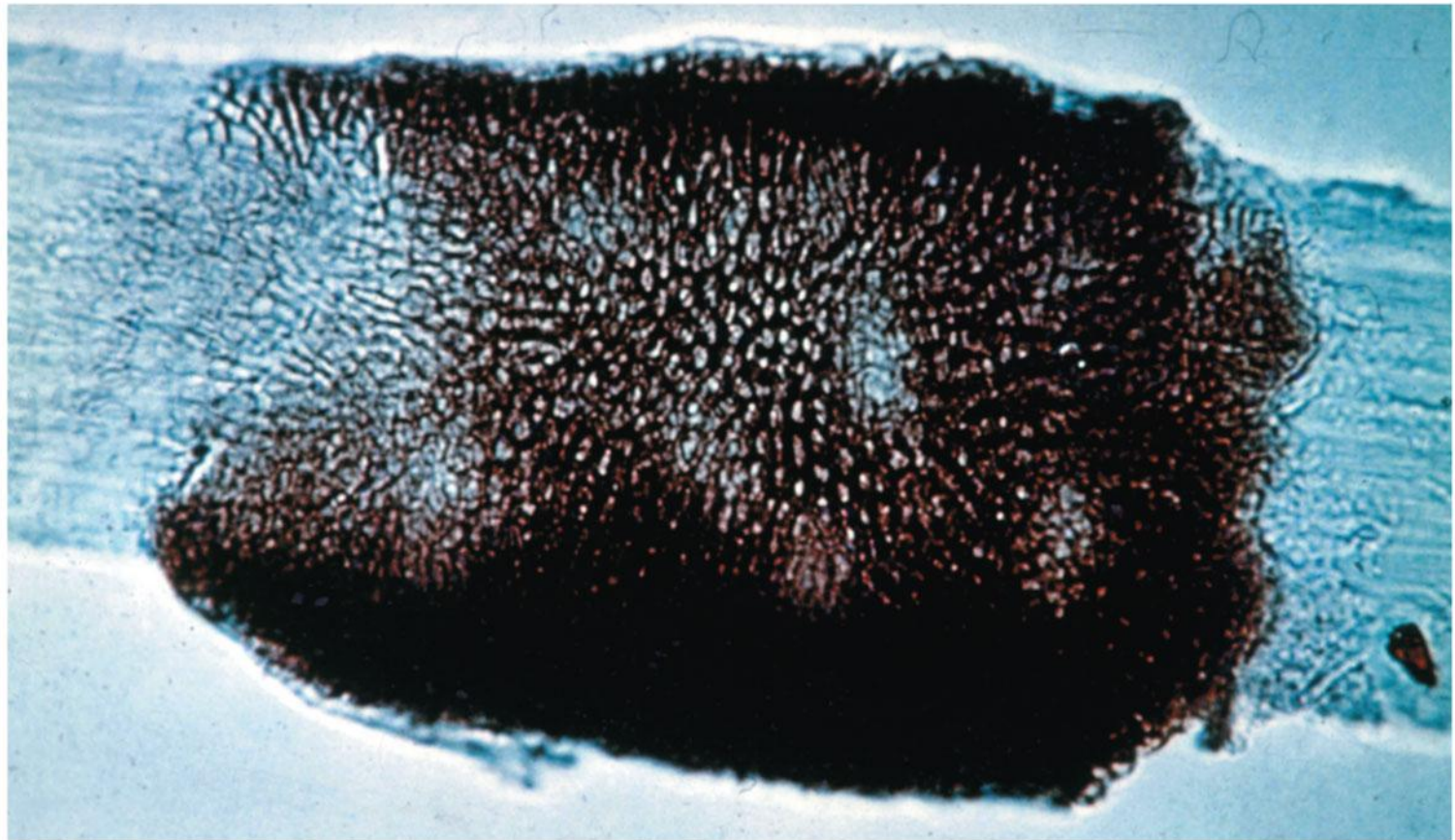
– Signs and symptoms

- Piedra
 - Irregular nodules on the hair shaft
- Pityriasis versicolor
 - Hypo- or hyperpigmented patches of scaly skin

– Pathogens and virulence factors

- *Piedraia hortae* causes black piedra
- *Trichosporon beigelii* causes white piedra
- Pityriasis caused by *Malassezia furfur*

Figure 19.17 Black piedra



LM

40 μ m

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Figure 19.18 Pityriasis versicolor



9/24/2026

Mycoses of the Hair Nails And Skin

- **Superficial Mycoses**

- Pathogenesis and epidemiology

- Superficial fungi produce keratinase, which dissolves keratin
 - Fungi often transmitted via shared hair brushes and combs

- Diagnosis, treatment, and prevention

- Piedra diagnosed by appearance and treated by shaving infected hair
 - Pityriasis identified by green color under ultraviolet light and treated with topical or oral drugs

Mycoses of the Hair, Nails, And Skin

2/ Cutaneous Mycoses

- Also called **Dermatophytoses**, these common diseases are caused by a group of related fungi, called **dermatophytes**.
- Dermatophytes fall into three genera, each with many species:
 - **Trichophyton**,
 - **Epidermophyton**, and
 - **Microsporum**.
- A defining characteristic of the dermatophytes is the ability to use keratin as a source of nutrition.
- This ability allows them to infect keratinized tissues and structures by producing keratinase,
 - such as **skin**, **hair**, and **nails**.
 - Cell-mediated immune responses damage deeper tissues

- There is some specificity, however. Whereas all three genera attack the skin

The Dermatophyte Genera and Diseases

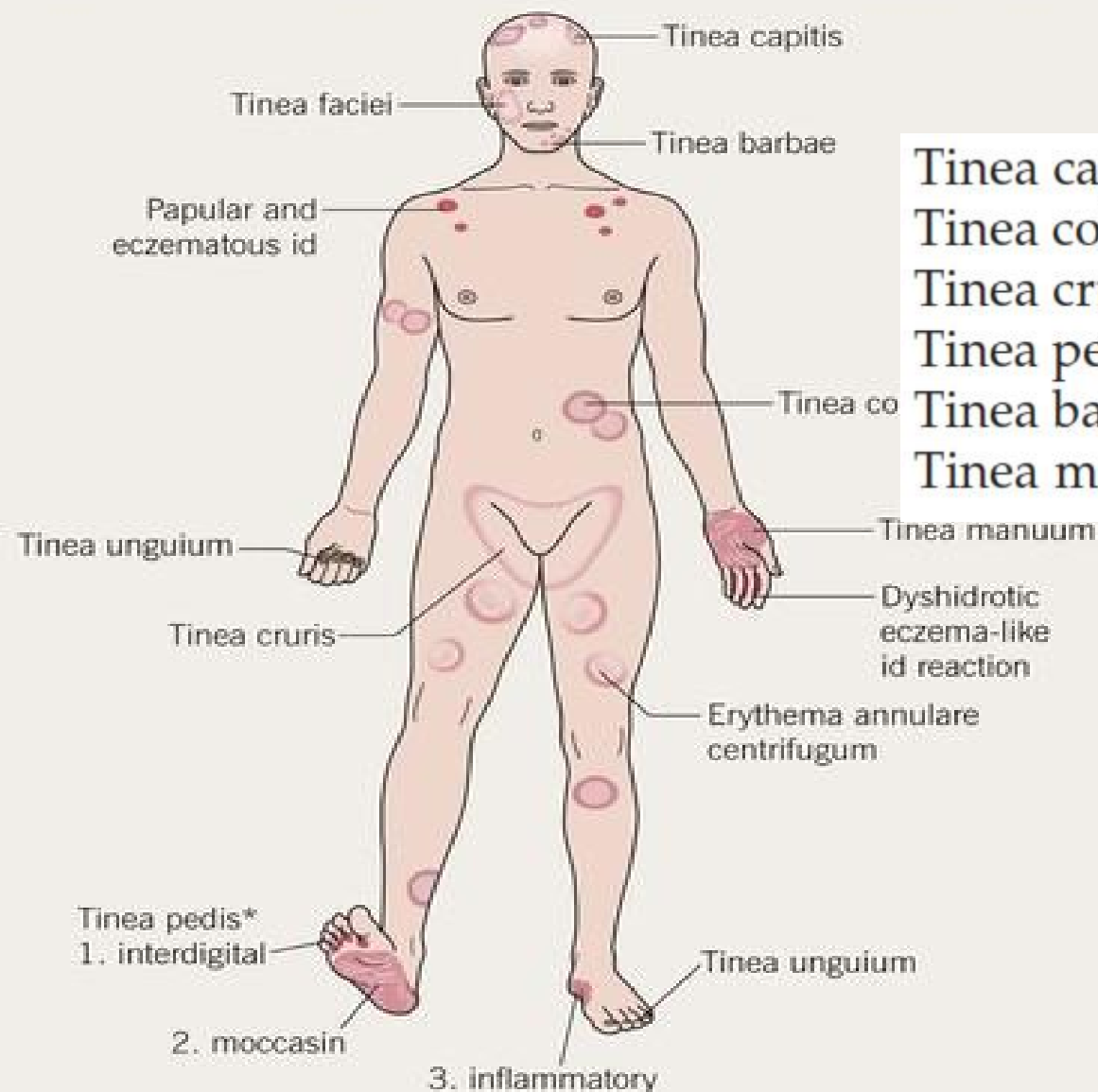
Genus	Name of Disease	Principal Targets	How Transmitted
<i>Trichophyton</i>	Ringworm of the scalp, body, beard, and nails Athlete's foot	Hair, skin, nails	Human to human, animal to human
<i>Microsporum</i>	Ringworm of scalp Ringworm of skin	Scalp hair Skin; not nails	Animal to human, soil to human, human to human
<i>Epidermophyton</i>	Ringworm of the groin and nails	Skin, nails; not hair	Strictly human to human

- On a worldwide scale, *T. rubrum* and *T. mentagrophytes* account for 80% to 90% of all dermatophytoses.

Clinical significance

- Dermatophytoses are characterized by ***itching***, *scaling skin patches* that can become inflamed and weeping.
- Specific diseases are usually identified according to affected tissue (for example, scalp, pubic area, or feet),
- Matched dermatophytoses disease with **affected area**
 - **Tinea pedis** (athlete's foot) -----foot
 - **Tinea corporis** (ringworm)-----any part of the skin
 - **Tinea cruris** (jock itch): -----pubic area
 - **Tinea capitis** (scalp ringworm)-----scalp
 - **Tinea unguium** (onychomycosis)-----nail

DERMATOPHYTE INFECTIONS OF THE SKIN AND POTENTIAL ASSOCIATED REACTIONS



Tinea capitis	: scalp
Tinea corporis	: body
Tinea cruris	: groin
Tinea pedis	: foot
Tinea barbae	: beard
Tinea manuum	: hand

Figure 19.19 Dermatophytosis (ringworm)



Figure 19.20 Athlete's foot



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Table 19.5 Common Dermatophytoses

19.5

Common Dermatophytoses

TABLE

Disease	Agents	Common Signs	Source
Tinea pedis ("athlete's foot")	<i>Trichophyton rubrum</i> ; <i>T. mentagrophytes</i> var. <i>interdigitale</i> ; <i>Epidermophyton floccosum</i>	Red, raised lesions on and around the toes and soles of the feet; webbing between the toes is heavily infected	Human reservoirs in toe webbing; carpeting holding infected skin cells
Tinea cruris ("jock itch")	<i>T. rubrum</i> ; <i>T. mentagrophytes</i> var. <i>interdigitale</i> ; <i>E. floccosum</i>	Red, raised lesions on and around the groin and buttocks	Usually spreads from the feet
Tinea unguium (onychomycosis)	<i>T. rubrum</i> ; <i>T. mentagrophytes</i> var. <i>interdigitale</i>	Superficial white onychomycosis: patches or pits on the nail surface; Invasive onychomycosis: yellowing and thickening of the distal nail plate, often leading to loss of the nail	Humans
Tinea corporis	<i>T. rubrum</i> ; <i>Microsporum gypseum</i> ; <i>M. canis</i>	Red, raised, ringlike lesions occurring on various skin surfaces (tinea corporis on the trunk, tinea capitis on the scalp, tinea barbae of the beard)	Can spread from other body sites; can be acquired following contact with contaminated soil or animals
Tinea capitis	<i>M. canis</i> ; <i>M. gypseum</i> ; <i>T. equinum</i> ; <i>T. verrucosum</i> ; <i>T. tonsurans</i> ; <i>T. violaceum</i> ; <i>T. schoenleinii</i>	Ectothrix invasion: fungus develops arthroconidia on the outside of the hair shafts, destroying the cuticle; Endothrix invasion: fungus develops arthroconidia inside the hair shaft without destruction; Favus: crusts form on the scalp, with associated hair loss	Humans; can be acquired following contact with contaminated soil or animals

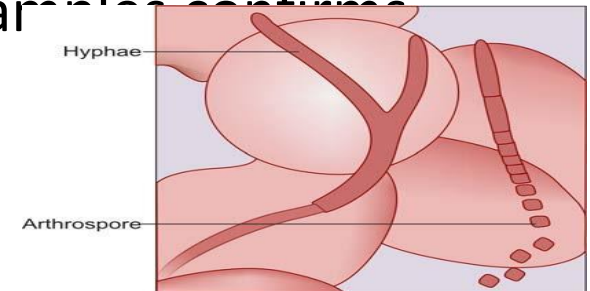
Mycoses of the Hair Nails And Skin

- **Cutaneous Mycoses...**

- **Diagnosis,**

- Diagnosed by clinical observation

- **KOH** preparation of skin or nail scrapings confirms diagnosis



- Direct microscopy (KOH) of nail/skin scrapings revealed

TREATMENT	Dermatophytoses
• Oral terbinafine or itraconazole are the drugs of choice for treatment of dermatophytosis. Duration of treatment depends on the affected site (1–2 weeks for skin lesions, 6 weeks for hair infection, 3 months for onychomycosis). They can be given as pulse therapy. • Alternate: Oral griseofulvin and ketoconazole may be given. • Topical lotion such as whitfield ointment or tolnaftate can be applied.	

Mycoses of the Hair, Nails, and Skin

3/ Sub-cutaneous mycoses (Wound Mycoses)

- Some fungi grow in deep tissues but do not become systemic
- Disease caused by certain types of fungi found in soil
- Fungi may enter the body through a Prick wounds and scrapes caused by twigs, thorns, or leaves
- After proper inoculation, fungi grow on deeper tissues in the body, including the hypodermis and bone
- It produce the **granulomatous** (tumour like) lesions in the skin, fascia (lining of muscles), and bones of the hands or feet
- they do not become systemic; that is, they do not spread throughout the body's systems.
- The principal subcutaneous mycoses are
 - A. **Chromomycosis,**
 - B. **Mycetoma,**
 - C. **Sporotrichosis and**

Mycoses of the Hair Nails and Skin

- **Wound Mycoses...**

- A/ Chromomycosis**

- Also called **chromoblastomycosis**
 - Multiple plaques on the lower leg.
 - characterized by **crusted warty nodules** that spread slowly along the lymphatics and develop crusty abscesses
 - Pathogens causing this mycosis include several species of pigmented soil fungi, for example, ***Phialophora*** and ***Cladosporium***
 - **Diagnosis:** Microscopic examination (10% KOH):
Clinical material: Skin scrapping/ Biopsy





Lesions of chromoblastomycosis may depict clinical polymorphism and elicit several differential diagnoses. The initial lesion of chromoblastomycosis (**1A**), may evolve to five main clinical types: nodular lesions on the lower leg (**1B**), verrucous lesion of the foot (**1C**), scar lesions on the knee and lower leg (**1D**), plaque lesion on the buttocks (**1E**), tumoral (cauliflower) lesions on the foot (**1G**) and mixed lesions composed by plaque, nodular and verrucous lesions involving the lower limb (**1H**).

Phaeohyphomycosis

- Is sub type of **Chromomycosis**
- Caused by over 30 genera of fungi
- Acquired when spores enter wounds
- *Exophiala jeanselmei* and *Wangiella dermatitidis* are the most common agents

- **Diagnosis:** Microscopic

TREATMENT

It consists of surgical removal (cryosurgery or laser therapy) of the lesion followed by antifungals (itraconazole).



Chromoblastomycosis

Mycoses of the Hair Nails and Skin

- Wound Mycoses

B/Mycetomas

- Also known as **Maduramycosis** or **Madura foot**
- Mycetoma appears as a localized abscess, usually on the feet, that discharges pus, serum, and blood through sinuses
- Most common causative agents are *Madurella grisea* and *Actinomadura madurae*.
- Defining characteristic of mycetoma is the presence of colored grains, composed of compacted hyphae, in the exudate.
- The color of the grains (**black, white, red, or yellow**) is characteristic of the causative organism and, therefore, useful in identifying the particular pathogen
- There is no effective chemotherapy for fungal mycetoma; the treatment is usually surgical excision.



C/ Sporotrichosis

- Caused by a dimorphic *Sporothrix schenckii*, a saprophyte in nature
- Subcutaneous infection usually limited to the arms and legs
- Occurs as fixed cutaneous or lymphocutaneous
- Also called **Rose handler's disease**
- Sporotrichosis is a chronic subcutaneous pyogranulomatous disease



NB

- Subcutaneous fungal infections are almost always acquired through *traumatic lacerations or puncture wounds*.
- The subcutaneous mycoses are not *transmissible from human to human* under ordinary conditions.

Parasitic Infestations of the Skin

- **Leishmaniasis** : is a disease of the skin and oral mucous membrane caused by an intracellular parasitic protozoan
 - **Signs and symptoms**
 - **Cutaneous**: Produces large painless skin lesions
 - **Mucocutaneous**: Skin lesions enlarge to encompass mucous membranes
 - **Visceral**: Parasite is spread by macrophages throughout body
 - **Pathogen and virulence factors**
 - *Leishmania* is the causative agent
 - Protozoan transmitted to humans by **female sand flies**

Figure 19.24 Mucocutaneous leishmaniasis



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Parasitic Infestations of the Skin

- **Leishmaniasis**

- **Pathogenesis and epidemiology**

- Infected macrophages stimulate inflammatory responses
 - Leishmaniasis endemic in parts of the tropics and subtropics

- **Diagnosis, treatment, and prevention**

- Diagnosed by microscopic identification of the protozoa
 - Most cases heal without treatment
 - Antimicrobials are needed for severe infections
 - Prevention involves reducing exposure to the reservoir host

Parasitic Infestations of the Skin

- **Infestations**

- Scabies
- Pediculosis
- Cutaneous larva migrans

Parasitic Infestations of the Skin

- **Scabies**

- Characterized by intense itching and rash at infestation site
- The mite *Sarcoptes scabiei* is the causative agent
- **Transmitted** via prolonged bodily contact
- Epidemics occur among people in crowded conditions
- Treated with mite-killing lotions and cleaning of contaminated items
- Prevented only by good personal hygiene



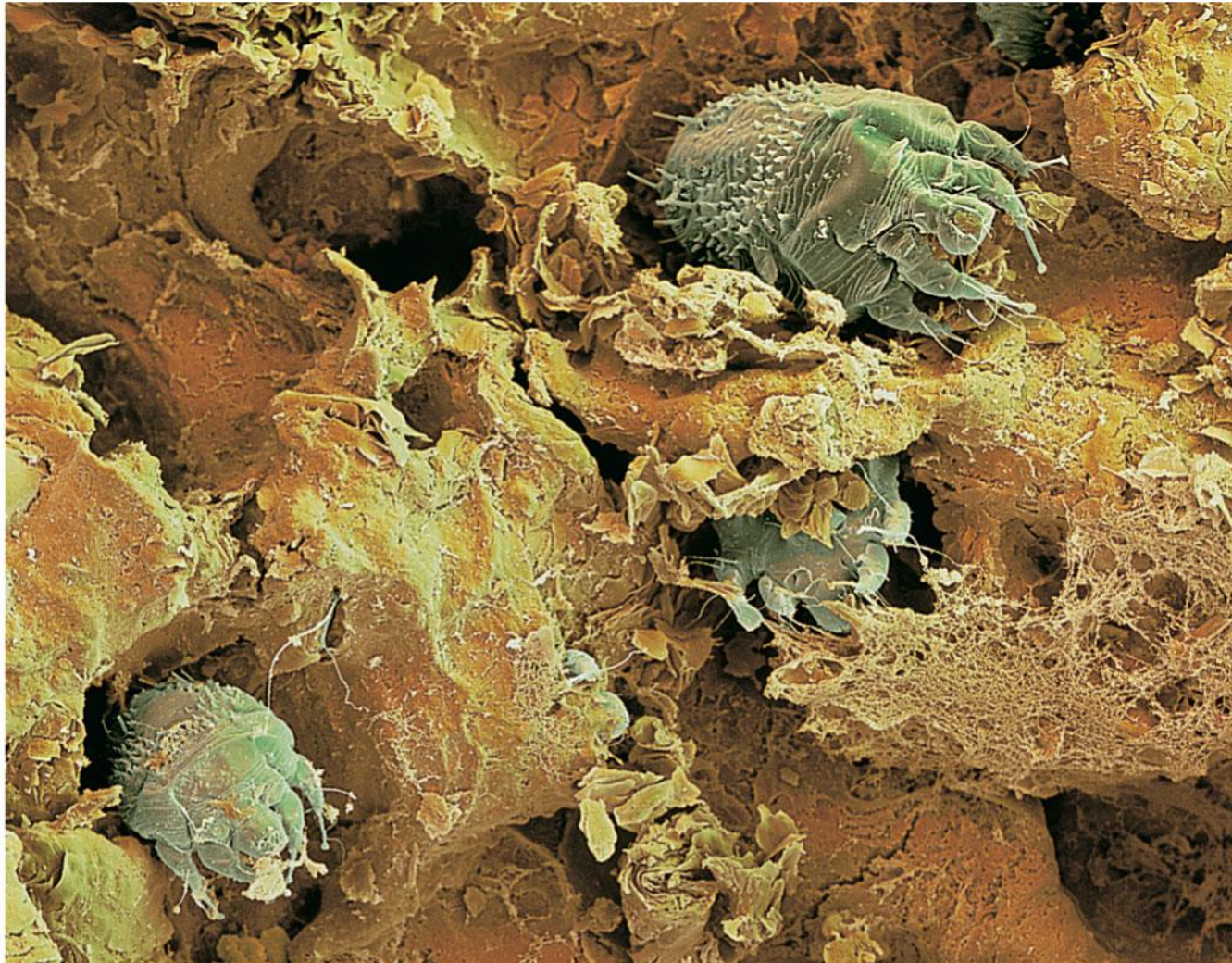
Parasitic Infestations of the Skin

Pediculosis: Head lice

- Common
- Louse feeds on scalp blood
- Nits on hair
- May be relatively
- asymptomatic



Figure 19.25 Scabies mites



SEM

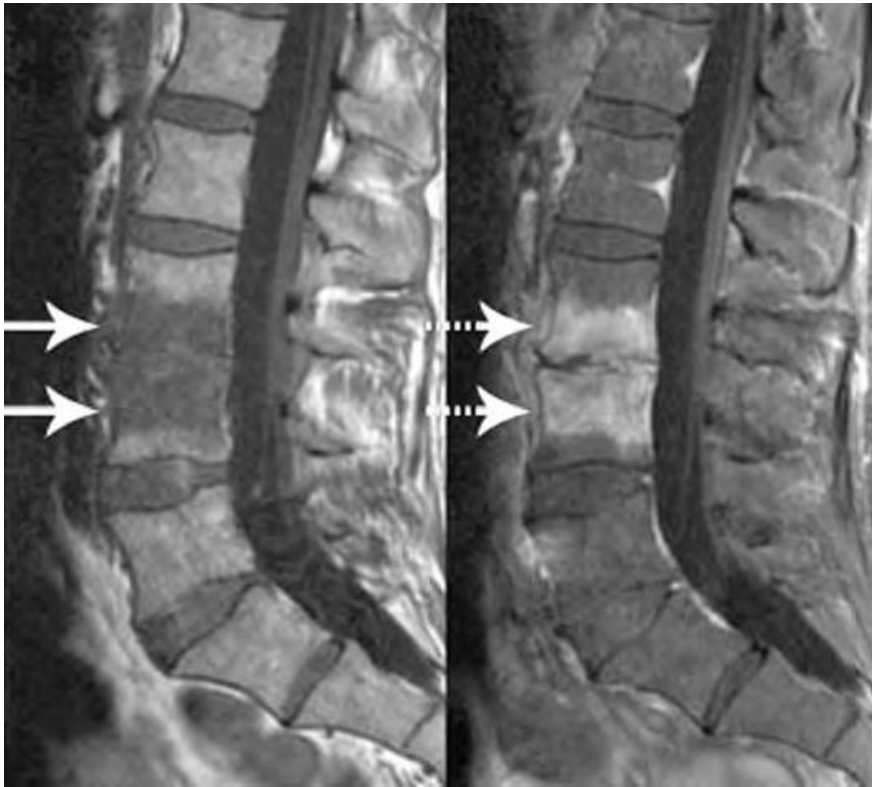
100 μm

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General Summary

The common cause of microbial Skin disease

- Bacterial skin infection
- Fungus/yeasts
- Viral including warts
- Infestations



BONES AND JOINTS

Part-2

INFECTIONS OF THE

Objective

At the end of this session the learner should be able to:

- Define osteomyelitis and septic arthritis
- Identify various etiological agents of bone and joint infection
- Describe epidemiology and its manifestation
- Describe the pathogenesis of various types of bone and joint infections
- Describe how to diagnosis and treatment

- Two different infectious diseases will be discussed in this section:

INFECTIONS OF THE BONES AND JOINTS...

1. Osteomyelitis and
2. Septic arthritis.

1- Osteomyelitis

- Is a progressive infection that can include one or multiple parts of the bone: e.g.,
 - periosteum,
 - medullary cavity, and
 - cortical bone
- Untreated, case may progress from inflammatory destruction of bone to -
 - necrosis (sequestra), followed by
 - new bone formation (involucrum).

Osteomyelitis ...

ETIOLOGY

- Osteomyelitis is usually a bacterial infection.
- *Most commonly caused by Staphylococcus aureus*
- In addition various types of organisms could causes osteomyelitis,
 - based on age group as indicated in the following table...

TABLE 32-1. Osteomyelitis: Type, Patient Profile, and Common Causes

Type of Osteomyelitis/Patient Profile	Common Causes
Hematogenous	Usually only one organism
Infants (<1 year)	<i>Staphylococcus aureus</i> , <i>Streptococcus agalactiae</i> (group B <i>Streptococcus</i>), <i>Escherichia coli</i>
Children (1–16 years)	<i>S aureus</i> , <i>Streptococcus pyogenes</i> (group A <i>Streptococcus</i>), <i>Haemophilus influenzae</i>
Persons >16 years	<i>S aureus</i> , coagulase-negative staphylococci (e.g., <i>Staphylococcus epidermidis</i>), gram-negative rod-shaped bacteria (e.g., <i>E coli</i> , <i>Pseudomonas</i> , <i>Serratia</i>)
Contiguous spread*	More likely to be polymicrobial
Microbiology depends on the primary site of infection	<i>S aureus</i> , <i>S pyogenes</i> (group A <i>Streptococcus</i>), <i>Enterococcus</i> (i.e., group D; <i>E faecalis</i> and <i>E faecium</i>), coagulase-negative staphylococci (e.g., <i>Staphylococcus epidermidis</i>), gram-negative rod-shaped bacteria (e.g., <i>Escherichia coli</i> , <i>Pseudomonas</i> , <i>Serratia</i>), anaerobes (e.g., <i>Prevotella</i> , <i>Bacteroides</i> , <i>Fusobacterium</i> , <i>Peptostreptococcus</i>)
Diabetic foot	<i>S aureus</i> , <i>Streptococcus</i> , <i>Enterococcus</i> , gram-negative rod-shaped bacteria (e.g., <i>Proteus mirabilis</i> , <i>Pseudomonas</i>), anaerobes (e.g., <i>Prevotella</i> , <i>Bacteroides</i> , <i>Fusobacterium</i> , <i>Peptostreptococcus</i>)

*Contiguous spread is the spread of an infection from a nearby site to the bone. Diabetic foot osteomyelitis is an example of contiguous spread; however, because of the high incidence of this infection, it is mentioned separately.

Osteomyelitis ...

CILINICAL MANIFESTATIONS

- The onset of symptoms of acute osteomyelitis can occur within 1–2 days, or chronic osteomyelitis can take weeks to months to develop

Among children

- more likely to develop acute long bone osteomyelitis, with symptoms of
 - *chills,*
 - *fever, and*
 - *malaise.*
 - *pain*
 - *localized swelling and*
 - *redness over the site of infection in the bone*

Among elders

- more likely to develop sub-acute or chronic vertebral osteomyelitis, and usually present with
 - localized lower back pain
 - tenderness with fever.
- **In chronic osteomyelitis**
 - localized pain may come and go with out fever.

Osteomyelitis ...

EPIDEMIOLOGY

- **Acute hematogenous osteomyelitis**
 - most commonly occurs in children
 - usually results in a single site of infection (e.g., tibia, femur, and humerus)
- **Osteomyelitis in adults**
 - usually involves the vertebral bodies.
- The overall prevalence of osteomyelitis is 1 per 5000 children, and
- About 50% of cases occur in preschool-aged children.
- The prevalence of osteomyelitis in neonates is 1 per 1000.
- **Spinal osteomyelitis** is more common in persons older than age 45 than in younger persons.

PATHOGENESIS

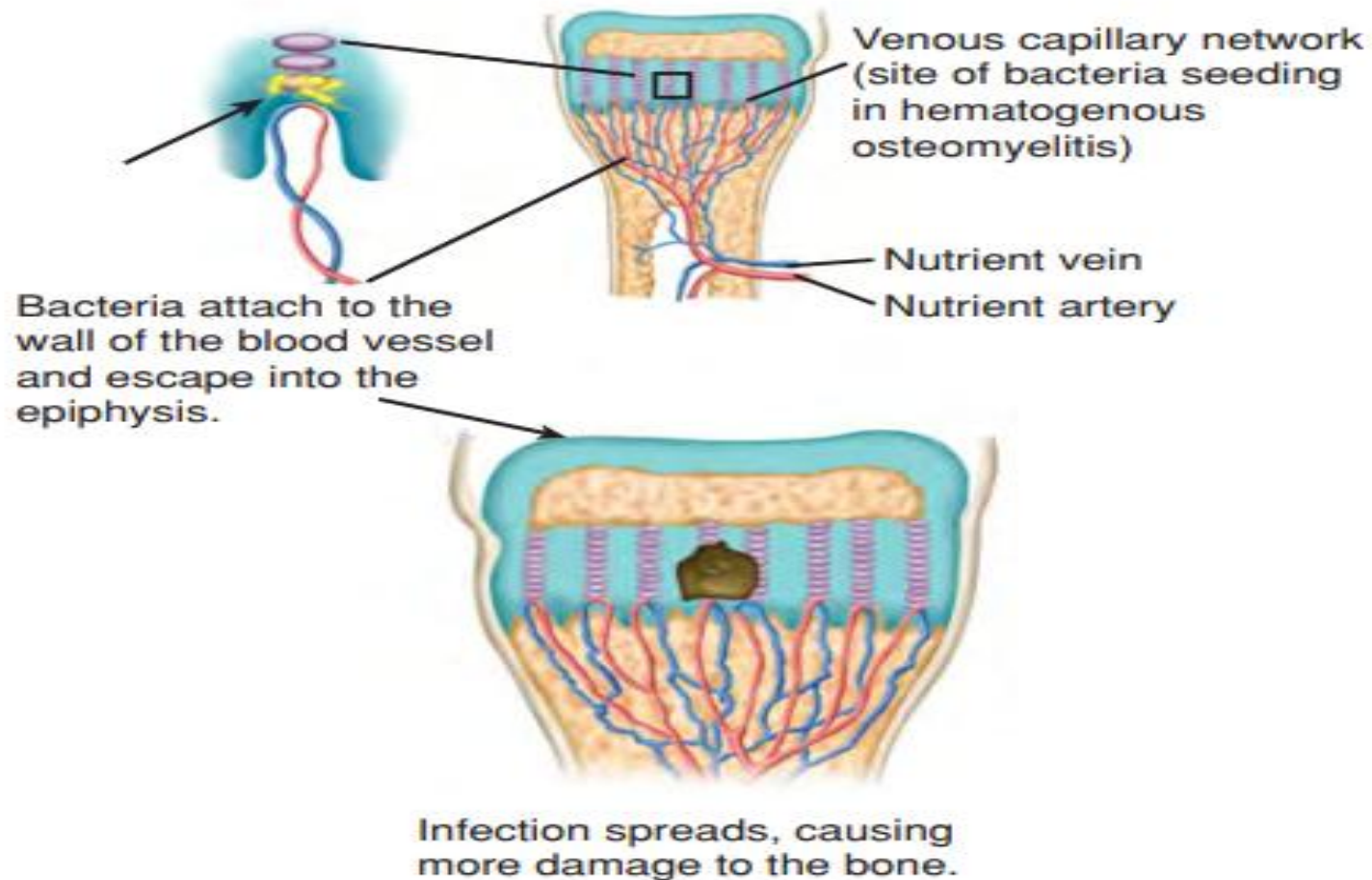


Figure 32-1. A schematic of the epiphysis of the long bone of a child. Note the site of bacterial seeding in the small looping blood vessels that supply nutrients to the epiphysis.

DIAGNOS

- Blood morphology
- Erythrocyte sedimentation rate (ESR) (high in both)
- C-reactive protein elevated in both acute & chronic
- Imaging studies
- Culture of sample from infected site



Figure 31-1. A radiograph of bone with osteomyelitis due to *Nocardia asteroides*. Note the moth-eaten appearance of the bone (red arrows) and the new bone growth over an area of dead bone (white arrow). Image courtesy of the Centers for Disease Control and Prevention, Dr Libero Ajello.

Deep tissue samples are NOT obtained to diagnose long bone infections in children. The epiphyseal plate can be damaged during the sampling, resulting in impaired bone growth. Instead, blood cultures should be obtained and, if negative, treatment is usually empirical.

TREATMENT AND PREVENTION

- Nafcillin or Oxacillin : for methicillin-sensitive *S aureus*
- vancomycin : for MRSA
- Penicillin G : for *Streptococcus* infections
- Ciprofloxacin: in gram-negative, rod-shaped bacterial infections
- Piperacillin-tazobactam and gentamicin: for *Serratia* or *Pseudomonas*
- Clindamycin or metronidazole: for anaerobic bacterial infection

2- SEPTIC ARTHRITIS

- Septic arthritis is an infection of the surface of the cartilage that lines the joint and
- The synovial fluid that lubricates the joint.
- As in osteomyelitis; Children and elderly adults are more likely to contract septic arthritis.

2- SEPTIC ARTHRITIS ...

ETIOLOGY

- **Polymicrobial**: *Viruses, fungi, and bacteria* can all cause infectious arthritis.
- However, bacterial infectious arthritis causes the most injury.
- There are two major classes of septic arthritis: **gonococcal and nongonococcal arthritis**.
- ***S aureus*** is the most common cause of **nongonococcal arthritis**

CILINICAL MANIFESTATIONS

- *triad of fever,*
- *joint pain, and*
- *impaired range of motion*
- swelling in a **single** joint

*nongonococcal septic
arthritis*

- Tenosynovitis,
- Dermatitis, and
- **Polyarthritis** syndrome
- Fever & malaise,

**Gonococcal
arthritis**



Figure 31-2. A photograph of a patient with gonococcal septic arthritis of the right elbow. Notice the swelling of the patient's arm. Image courtesy of the Centers for Disease Control and Prevention.

EPIDEMIOLOGY

- Most cases occur in the very young and the very old and among intravenous drug users.
- About 45% of cases are persons older than 65 years of age.
- Mortality rates depend on the agent causing the arthritis.
- In **septic arthritis** infection due to *S aureus*, the mortality rate can be as high as 50%.
- Nearly 50% of patients who develop septic arthritis have an underlying chronic joint disease (e.g., **rheumatoid** arthritis, **osteoarthritis**).

In adults,

- **knee (most affected)**
- **hip,**
- **shoulder,**
- **ankle, and**
- **wrists;**

In children

- **hip** (most affected, followed by
- the **knee**.

PATHOGENESIS

- **Organisms** can enter the joint by:
 - direct inoculation,
 - contiguous spread from infected periarticular tissue.
- However, ***the most common route of infection is following bacteremia.***
 - » *Bacteremia* through (UTI, intravenous drug use, intravenous catheters, endocarditis, and soft tissue infections)
- Following bacteremia; some bacteria have surface factors that promote their adherence to the joint.

PATHOGENESIS...

- Bacterial growth in the joint causes an acute inflammatory reaction.
- Results in infiltration of polymorphonuclear leukocytes.
- Injury to joint cartilage is due to the synthesis of **cytokines** and **inflammatory** products and
- Bacterial production of factors such as **chondrocyte proteases** of ***S aureus***, which cause joint damage.

DIAGNOSIS

❖ Laboratory test

- white blood cell count,
- gram stain smear, and
- culture of the synovial fluid are essential diagnosis
- Blood cultures should also be used
- In cases of gonococcal septic arthritis,
 - pharyngeal,
 - rectal,
 - cervical, or
 - urethral specimens

TABLE 32-2. Synovial Fluid Analysis to Differentiate Septic Arthritis from Noninflammatory and Inflammatory Arthritis

Laboratory Test	Normal Synovial Fluid	Septic Arthritis	Noninflammatory Arthritis	Inflammatory Arthritis
Clarity and color	Clear	Opaque, yellow to green	Clear, yellow	Translucent, yellow, or opalescent
Viscosity	High	Variable	High	Low
White blood cells/mm ³	<200	>100,000*	200–2000	2000–10,000
Percent polymorphonuclear leukocytes	<25%	>75%	<25%	≥50%
Total protein g/dL	1–2	3–5	1–3	3–5
Glucose concentration (concentration in joint relative to concentration in blood)	Nearly equal	<25%	Nearly equal	50–80%
Culture	Negative	Often positive in nongonococcal arthritis; usually negative in gonococcal arthritis	Negative	Negative
Disease or condition	Normal	Septic arthritis	Osteoarthritis, trauma to joint	Rheumatoid arthritis Reiter disease Partially treated septic joint Fungal or viral infection Gout or pseudogout Acute rheumatic fever

*Numbers may be lower in partially treated joint infections and in joint infections due to low virulence organisms.

TREATMENT AND PREVENTION

Nongonococcal septic arthritis treatment

- involves two essential components.
 - 1st , ***draining and washing of purulent exudate should be completely done by arthroscopy or surgery***
 - 2nd providing ***appropriate antibiotic***

Gonococcal septic arthritis

- Also requires ***complete drainage and washing of the purulent synovial fluid from the joint and antibiotic therapy with intravenous ceftriaxone***
- after clinical improvement. Oral
 - cefixime,
 - ciprofloxacin,
 - ofloxacin, or
 - levofloxacin should be used
- **Prevention of septic arthritis:** Avoiding trauma and unsafe sex

The End
Thank you!!
Any ???